ENGLISH FOR WORK / ENGLISH FOR BUILDING COMFORT AND REFRIGERATION TEAMS

HVAC & Refrigeration English

Clear readings. Reliable handovers.



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Eight practical lessons. Eleven extended conversations.
Vocabulary, phrase banks, guided speaking, and explained answers.

Revised 2026 edition / Upper-intermediate to advanced

Your route through the book

01 Clarifying comfort complaints and thermostat readings, page 4

Separate a thermostat setpoint from a measured room temperature while preserving the time and location of a comfort complaint.

02 Identifying equipment and linked components, page 14

Reconcile conflicting equipment references using existing records and preserve an unverified relationship between components.

03 Explaining airflow and distribution reports, page 26

Translate supply and return labels into plain language while preserving the difference between a comfort report and an airflow measurement.

04 Checking equipment ratings and refrigerant references, page 36

Compare like quantities, request missing equipment data, and distinguish refrigerant identification from approval to select or service a system.

05 Reporting moisture, symptoms, and service findings, page 50

Reconcile a historical report with a later observation and avoid treating an earlier maintenance concern as a proven cause.

06 Discussing refrigeration alarms and temperature records, page 60

Report an alarm chronology accurately, distinguish cabinet and product information, and route a food-safety question to the responsible lead.

07 Coordinating ductwork, access, and changes, page 73

Acknowledge an appearance preference, explain unassessed consequences, and present two bounded choices without promising price or timing.

08 Handing over commissioning and maintenance information, page 85

Separate document receipt from completed verification and distinguish assigned correction work from an unassigned scheduling responsibility.

Professional communication notes, page 3

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Suggested study range: B2-C1. This is an editorial study range, not a certified proficiency assessment.

Keep the reading attached to its meaning.

Technical English in this field depends on what a number measures, which component a label identifies, and what a record actually establishes. Practice explaining those distinctions to colleagues and customers.

Name the value before interpreting it

A thermostat setpoint is a target, not automatically the measured room temperature. Cooling capacity and electrical input can both use kilowatts but describe different quantities. Keep the quantity name and unit together.
"The 22 degrees Celsius is the setpoint; no ambient reading has been supplied."

Preserve component identity

Similar suffixes on an outdoor unit and an air handler do not prove a verified pairing. Use the project register and confirmed references rather than treating a familiar shorthand label as the whole system.
"CU-4 and AH-4 appear in separate records; their relationship needs confirmation."

Separate history, findings, and decisions

A past moisture report can coexist with a dry floor during a later visit. A cabinet display and a defrost entry do not establish food temperatures or release food for service. Name the evidence and the responsible decision route.
"The display recorded 8 degrees Celsius; product temperatures and event duration are not supplied."

Keep unfinished handover items visible

Receiving a commissioning report does not complete a deferred check. Correct documents, scheduling responsibility, and update dates need separate entries. A named document owner does not automatically own every outstanding action.
"Nia owns document correction; responsibility for scheduling the seasonal check remains open."

Professional boundaries

The projects, products, values, and records are fictional. This book teaches English, not equipment operation, diagnosis, installation, refrigerant handling, food-safety decisions, or compliance. Follow actual training, authorization, manufacturer instructions, site procedures, and applicable requirements. No dialogue authorizes roof access, testing, adjustment, refrigerant work, food release, or a system alteration. Equipment readings and document receipt do not establish safety or overall performance.

A Read the situation

LESSON 01 / CLARIFYING COMFORT COMPLAINTS AND THERMOSTAT READINGS

Separate a thermostat setpoint from a measured room temperature while preserving the time and location of a comfort complaint.

> **Twenty-two is the target, not the finding**
>
> Office manager Asha tells service coordinator Ben that the meeting room is 22 degrees Celsius. The number is actually the thermostat setpoint, not a confirmed ambient-temperature reading. Staff report feeling warm there after 14:00. No cause or equipment fault has been established. Ben needs to clarify the value, keep the staff report attributed, and record the afternoon pattern. The conversation uses existing information and gives no instruction to adjust or test the equipment.

People: Asha, Office manager; Ben, Service coordinator.

Read the record

Choose one answer per question using the facts above.

1. What does 22 degrees Celsius represent?

A. An outdoor temperature
B. The thermostat setpoint
C. A verified ambient-temperature reading
D. A supply-air measurement

2. When do staff report feeling warm?

A. Exactly at a confirmed failure time
B. Throughout every day without exception
C. After 14:00
D. Only before 08:00

3. What is established about the equipment?

A. The system is confirmed fault-free
B. No cause or fault has been established
C. The thermostat sensor has failed
D. The cooling capacity is inadequate

Answers and explanations, page 113

B Vocabulary in use

CORE TERMS

HVAC
Heating, ventilation, and air conditioning.

discuss the HVAC system

thermostat
Control associated with maintaining a selected temperature.

identify the thermostat

setpoint
Target value used by a control system.

distinguish the setpoint

ambient temperature
Temperature of the surrounding air at the relevant location.

record ambient temperature

temperature reading
Value reported by a temperature-measuring device.

identify the temperature reading

degrees Celsius
Temperature unit used in the supplied report.

state the value in degrees Celsius

comfort complaint
Occupant report that environmental conditions feel uncomfortable.

log a comfort complaint

occupant feedback
Account supplied by people using the space.

preserve occupant feedback

sensible cooling
Heat removal associated with lowering air temperature without counting moisture condensation.

distinguish sensible cooling

afternoon pattern
Reported tendency occurring later in the day.

record the afternoon pattern

sensor
Device responding to a physical condition for measurement or control.

identify the relevant sensor

measured value
Quantity established by a measurement.

distinguish the measured value

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

HVAC HVAC

discuss the HVAC system

Keep the meaning precise: Heating, ventilation, and air conditioning.

degrees Celsius DEGREES CELSIUS

state the value in degrees Celsius

Keep the meaning precise: Temperature unit used in the supplied report.

B Precision and professional shorthand

EXTEND YOUR RANGE

displayed value
Number shown by a device or interface.
clarify the displayed value

latent cooling
Heat removal associated here with condensing water vapor from air.
account for latent cooling

occupancy
Use of a space by people.
describe room occupancy

zone
Defined area treated as a control or service region.
identify the relevant zone

equipment fault
Established malfunction of a system component.
avoid assuming an equipment fault

service description
Concise account used to request relevant attention.
clarify the service description

target temperature
Temperature the control is intended to seek.
state the target temperature

relative humidity
Water-vapor level relative to saturation at the same temperature.
report relative humidity

control schedule
Timing arrangement for control settings or operation.
review the control schedule

heat gain
Heat added to a space from its surroundings or internal sources.
assess heat gain

reported period
Time span described in an account.
preserve the reported period

attributed report
Statement linked to the person or group who supplied it.
retain an attributed report

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

measured value MEASURED VALUE

distinguish the measured value

Keep the meaning precise: Quantity established by a measurement.

occupancy OCCUPANCY

describe room occupancy

Keep the meaning precise: Use of a space by people.

C Phrases that move the work forward

Ask what the number means

Is 22 the setpoint or a measured room temperature?

Preserve the missing reading

No confirmed ambient reading has been supplied.

Keep the time pattern

The reports concern the period after 14:00.

Separate the evidence

We have a setpoint and an occupant report, not a diagnosis.

Ask about existing information

What does the current record identify that value as?

Use the correct unit

The value is stated in degrees Celsius.

Retain the account

I will record what the staff have reported.

Read back the distinction

Setpoint 22; staff feel warm after 14:00; ambient reading unavailable.

Name the target

The thermostat is set to 22 degrees Celsius.

Acknowledge the people

Staff report feeling warm in the meeting room.

Avoid a contradiction

A target setting does not disprove a comfort complaint.

Avoid assigning a failed part

No sensor fault has been established.

Keep the location

Please keep meeting room in the service description.

Avoid prescribing an adjustment

This clarification does not specify a new setting.

Avoid inventing a measured excess

We do not have a measured temperature above the target.

Close with an accurate request

I will pass on that description for the relevant assessment.

Precision in real exchanges

Purpose: Acknowledge the people
Useful wording: Staff report feeling warm in the meeting room.

Purpose: Keep the time pattern
Useful wording: The reports concern the period after 14:00.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which wording accurately uses the number?

A. The thermostat setpoint is 22 degrees Celsius
B. The room was measured at exactly 22
C. Supply air is confirmed at 22
D. The outdoor temperature caused the problem at 22

2. Which summary preserves the complaint?

A. A sensor fault begins at 14:00
B. Staff feel warm in the meeting room after 14:00; ambient reading not supplied
C. Staff are mistaken because the setting is 22
D. Every room overheats all day

3. A colleague needs the term for this meaning: Temperature of the surrounding air at the relevant location. Which term fits?

A. latent cooling
B. equipment fault
C. ambient temperature
D. sensible cooling

D Choose the response that fits / continued

4. Rebuild the exchange from "Twenty-two is the target, not the finding". Asha: "The reports concern the meeting room after two in the afternoon. I don't have the same complaint about every room or every part of the day." Ben: [missing response] Asha: "Will the service team dismiss the complaint if the request says the thermostat is set to twenty-two? The people still feel uncomfortable." Which response belongs in the gap?

A. Certainly. I will state degrees Celsius explicitly. A number without its unit and its meaning can become misleading when the message is forwarded.

B. It is. No equipment fault has been established. We should not name a failed sensor or another component simply to give the complaint a technical-sounding cause.

C. Before I record it, is twenty-two the setpoint, or is it a separate measured room-temperature value? Those mean different things in a service request.

D. That is a useful afternoon pattern. I will retain the meeting-room location and after 14:00, so the request does not become a whole-office complaint.

Answers and explanations, page 113

One useful distinction

Twenty-two degrees Celsius is the setpoint in this case. It must not be relabeled as ambient temperature. The complaint concerns staff feeling warm in the meeting room after 14:00, not a verified temperature or fault.

E Conversation

Twenty-two is the target, not the finding

Twenty-two is the target, not the finding

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

setpoint / equipment fault / target temperature / degrees Celsius / attributed report / ambient temperature / temperature reading / occupant feedback / afternoon pattern / service description

01 **Asha:** Ben, the meeting room is twenty-two degrees, but staff keep saying it feels warm after lunch. How should I describe that without sounding contradictory?

02 **Ben:** Before I record it, is twenty-two the [1] _____, or is it a separate measured room-temperature value? Those mean different things in a service request.

03 **Asha:** It's the thermostat setting, not a separate room measurement. I copied the number from the information the office team gave me.

04 **Ben:** Then we should not call it the [2] _____. We can record the target and the staff account without treating the setting as proof of the room condition.

05 **Asha:** The reports concern the meeting room after two in the afternoon. I don't have the same complaint about every room or every part of the day.

06 **Ben:** That is a useful [3] _____. I will retain the meeting-room location and after 14:00, so the request does not become a whole-office complaint.

07 **Asha:** Will the service team dismiss the complaint if the request says the thermostat is set to twenty-two? The people still feel uncomfortable.

08 **Ben:** No. We can preserve the [4] _____ and clarify the number at the same time. A target does not establish what everyone in the room actually experiences.

09 **Asha:** I had written possible sensor failure to make the request more specific. But the information doesn't establish a failed sensor, does it?

10 **Ben:** It is. No [5] _____ has been established. We should not name a failed sensor or another component simply to give the complaint a technical-sounding cause.

Conversation continues on the next page.

Twenty-two is the target, not the finding

CONVERSATION / CONTINUED

11 **Asha:** Please include Celsius. Our overseas colleagues sometimes use Fahrenheit, and I'd rather the unit didn't disappear when the request is forwarded.

12 **Ben:** Certainly. I will state [6] _____ explicitly. A number without its unit and its meaning can become misleading when the message is forwarded.

13 **Asha:** When we say no confirmed reading supplied, we mean this account lacks it. We don't know whether someone has measured the room on another occasion.

14 **Ben:** Yes. The missing [7] _____ is a limit of this information. It does not establish the history of every measurement or the condition of the equipment.

15 **Asha:** I'll keep the request to the existing information. I don't want my summary mistaken for an instruction to change the thermostat or test a component.

16 **Ben:** Agreed. We are clarifying a [8] _____, not issuing control instructions. Any assessment belongs with the relevant team and its actual procedures.

17 **Asha:** Could you read it back with the target and missing measurement separate? I need a short version for the facilities manager.

18 **Ben:** Here is the [9] _____: staff feel warm in the meeting room after 14:00; thermostat setpoint 22 degrees Celsius; no confirmed ambient reading supplied; cause unestablished.

19 **Asha:** That preserves the staff account without inventing a measured temperature above target. I'll stop describing twenty-two as the actual room condition.

20 **Ben:** I will pass on that record. It distinguishes the [10] _____ from the actual condition that still needs assessment, while keeping the staff's reported experience visible.

CHECK THE GAPS

1. setpoint / **2.** ambient temperature / **3.** afternoon pattern / **4.** occupant feedback / **5.** equipment fault / **6.** degrees Celsius / **7.** temperature reading / **8.** service description / **9.** attributed report / **10.** target temperature

Why these answers fit, page 114

F Check accuracy and scope

Twenty-two is the target, not the finding

Return to the case

Office manager Asha tells service coordinator Ben that the meeting room is 22 degrees Celsius. The number is actually the thermostat setpoint, not a confirmed ambient-temperature reading. Staff report feeling warm there after 14:00. No cause or equipment fault has been established. Ben needs to clarify the value, keep the staff report attributed, and record the afternoon pattern. The conversation uses existing information and gives no instruction to adjust or test the equipment.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What should Ben avoid inferring?

A. The reported room
B. The afternoon pattern
C. A particular equipment fault
D. The value's unit

2. Which question best clarifies the displayed value?

A. Does the record label 22 as the target or the measured room temperature?
B. Why is the compressor broken?
C. Which new setting should we choose immediately?
D. Can the client test the equipment now?

Answers and explanations, page 116

Listen for the distinction

Switch roles for turns 11-20. Keep Celsius with the target and distinguish missing information from no measurement ever taken.

Correct the number label without dismissing the people

A setpoint can be accurate while a comfort complaint is still meaningful. Ask what the displayed value represents before treating it as a measurement. Acknowledge the occupants and preserve their reported pattern without diagnosing a fault or prescribing a control change.

F Say it / transfer to a new case

Label the value correctly

Complete the service report using the target, time pattern, and missing measurement. Do not add a diagnosis.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

ambient / setpoint / 14:00 / unestablished

01 **Manager:** The thermostat [1] _____ is 22 degrees Celsius.

02 **Coordinator:** Staff feel warm after [2] _____.

03 **Manager:** No confirmed [3] _____ reading is supplied.

04 **Coordinator:** The cause remains [4] _____.

CHECK THE GAPS

1. setpoint / **2.** 14:00 / **3.** ambient / **4.** unestablished

Why these answers fit, page 116

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Twenty-two is the target, not the finding". Asha: "When we say no confirmed reading supplied, we mean this account lacks it. We don't know whether someone has measured the room on another occasion." Ben: [missing response] Asha: "I'll keep the request to the existing information. I don't want my summary mistaken for an instruction to change the thermostat or test a component." Which response belongs in the gap?

A. I will pass on that record. It distinguishes the target temperature from the actual condition that still needs assessment, while keeping the staff's reported experience visible.

B. Yes. The missing temperature reading is a limit of this information. It does not establish the history of every measurement or the condition of the equipment.

C. Here is the attributed report: staff feel warm in the meeting room after 14:00; thermostat setpoint 22 degrees Celsius; no confirmed ambient reading supplied; cause unestablished.

D. It is. No equipment fault has been established. We should not name a failed sensor or another component simply to give the complaint a technical-sounding cause.

Answers and explanations, page 117

A Read the situation

LESSON 02 / IDENTIFYING EQUIPMENT AND LINKED COMPONENTS

Reconcile conflicting equipment references using existing records and preserve an unverified relationship between components.

AC-4, CU-4, and AH-4 are not yet a pairing

A school service request calls the system AC-4. The roof plan labels an outdoor unit CU-4, while the indoor equipment register identifies air handler AH-4. Their relationship is not confirmed. Facilities manager Mira has the equipment register and discusses the request with service coordinator Owen. They need to preserve the three source references and seek confirmation through the records process. Neither person needs to access the roof or inspect equipment to clarify the existing documents.

People: Mira, Facilities manager; Owen, Service coordinator.

Read the record

Choose one answer per question using the facts above.

1. Which label appears in the service request?

A. AH-4 only
B. A verified combined asset number
C. AC-4
D. CU-4 only

2. What does the register identify as AH-4?

A. An indoor air handler
B. A confirmed outdoor condenser
C. A roof-access permit
D. A room thermostat setting

3. What is unresolved?

A. Whether Mira has the register
B. Whether the request uses AC-4
C. Whether the roof plan contains CU-4
D. The relationship between CU-4 and AH-4

Answers and explanations, page 118

B Vocabulary in use

CORE TERMS

equipment register
Record identifying equipment and relevant asset information.

consult the equipment register

asset tag
Assigned identifier used to distinguish an equipment item.

preserve the asset tag

system reference
Label used to identify an overall system.

clarify the system reference

component reference
Label identifying a particular part of a system.

retain the component reference

air handler
Equipment that moves and may condition air in a building system.

identify the air handler

outdoor unit
Equipment assembly located outside the served interior.

identify the outdoor unit

condensing unit
Assembly associated with refrigerant compression and heat rejection in relevant systems.

verify the condensing-unit reference

indoor unit
Equipment assembly located within the building or served space.

identify the indoor unit

equipment pairing
Confirmed relationship between components intended to operate together.

verify the equipment pairing

served area
Space associated with the service from an identified system.

confirm the served area

roof plan
Drawing showing relevant roof-level features and equipment.

cite the roof plan

record source
Document or system from which information was obtained.

identify the record source

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

equipment register EQUIPMENT REGISTER

consult the equipment register

Keep the meaning precise: Record identifying equipment and relevant asset information.

outdoor unit OUTDOOR UNIT

identify the outdoor unit

Keep the meaning precise: Equipment assembly located outside the served interior.

B Precision and professional shorthand

EXTEND YOUR RANGE

cross-reference

Link connecting corresponding entries in different records.

establish a cross-reference

label discrepancy

Difference between identifiers in related information.

record the label discrepancy

model number

Manufacturer identifier for a product design or version.

confirm the model number

serial number

Identifier distinguishing an individual manufactured unit.

record the serial number

nameplate

Manufacturer label carrying equipment identification or ratings.

use recorded nameplate information

asset hierarchy

Organized relationship between systems and their components.

clarify the asset hierarchy

service request

Record asking for attention to an identified issue or system.

reconcile the service request

records reconciliation

Process of resolving differences between documented information.

complete records reconciliation

unverified link

Relationship that has not been confirmed.

retain an unverified link

identification query

Request to clarify which equipment is meant.

raise an identification query

record custodian

Person responsible for maintaining or providing a record.

identify the record custodian

confirmed mapping

Verified correspondence between identifiers or components.

obtain confirmed mapping

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

record source RECORD SOURCE

identify the record source

Keep the meaning precise: Document or system from which information was obtained.

nameplate NAMEPLATE

use recorded nameplate information

Keep the meaning precise: Manufacturer label carrying equipment identification or ratings.

C Phrases that move the work forward

Quote the request
The service request calls the system AC-4.

Quote the plan
The roof plan labels an outdoor unit CU-4.

Quote the register
The indoor register identifies air handler AH-4.

Preserve source attribution
Each label should stay attached to its source.

Avoid a number shortcut
The shared four does not prove that the components are paired.

Name the unresolved link
The relationship between CU-4 and AH-4 is unverified.

Use existing records
You already have the equipment register.

Request reconciliation
We need a confirmed cross-reference between these entries.

Separate system and component
A system label may not be the identifier of one physical unit.

Avoid assumed renaming
I will not replace AC-4 with CU-4 by guesswork.

Keep the served area open
We have not confirmed the served area from these labels.

Distinguish identifiers
A model number and a serial number identify different things.

Keep the request precise
Please retain all three references in the query.

Avoid directing access
This is a document query, not a request for roof access.

Read back the gap
The records name the items but do not confirm their relationship.

Close with the needed result
We need confirmed mapping before merging the references.

Precision in real exchanges

Purpose: Preserve source attribution
Useful wording: Each label should stay attached to its source.

Purpose: Avoid a number shortcut
Useful wording: The shared four does not prove that the components are paired.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which summary is accurate?

A. All three labels identify one physical item

B. Request AC-4; roof plan CU-4; register AH-4; relationship unverified

C. AC-4 is definitely another name for CU-4

D. AH-4 and CU-4 are paired because both end in four

2. What should Owen ask Mira to provide?

A. A revised request replacing AC-4 with CU-4 immediately

B. A cross-reference based only on the shared numeral

C. A served-area label borrowed from another request

D. The relevant existing register information for reconciliation

3. A colleague needs the term for this meaning: Label identifying a particular part of a system. Which term fits?

A. unverified link

B. equipment pairing

C. component reference

D. model number

D Choose the response that fits / continued

4. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "They all end in four. I thought that would be enough to link the indoor and outdoor entries, but the records don't actually confirm it." Owen: [missing response] Mira: "I have the register here already. Can we resolve the question through the records process instead of asking someone to go onto the roof?" Which response belongs in the gap?

A. Then leave the equipment pairing unverified. The shared numeral suggests a question to resolve; it doesn't establish that those components operate together.

B. Yes. Send the relevant equipment register entries and roof-plan reference. We're reconciling existing documents, not requesting physical access or a new equipment inspection.

C. Ask for a confirmed cross-reference between request AC-4, plan item CU-4, and register item AH-4, including the component relationship and the served area.

D. Let's keep the record source beside each identifier. What does the roof plan call the outdoor equipment, and what does the indoor register say?

Answers and explanations, page 118

> **One useful distinction**
>
> AC-4 appears in the request, CU-4 on the roof plan, and AH-4 in the indoor register. Keep all three with their sources. A shared numeral does not confirm a component pairing or served area.

E Conversation

AC-4, CU-4, and AH-4 are not yet a pairing

AC-4, CU-4, and AH-4 are not yet a pairing

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

model number / record source / equipment pairing / system reference / service request / equipment register / served area / cross-reference / air handler / confirmed mapping

01 **Mira:** Owen, the school's request says AC-4, but I can't match that cleanly to the project documents. I don't want to send the wrong reference.

02 **Owen:** Let's keep the [1] _____ beside each identifier. What does the roof plan call the outdoor equipment, and what does the indoor register say?

03 **Mira:** The roof plan labels the outdoor unit CU-4. The indoor register lists AH-4, an air handler. Only the service request uses AC-4.

04 **Owen:** So the [2] _____ has its own reference. AC-4 may refer to the system more broadly; we shouldn't treat all three labels as one physical item.

05 **Mira:** They all end in four. I thought that would be enough to link the indoor and outdoor entries, but the records don't actually confirm it.

06 **Owen:** Then leave the [3] _____ unverified. The shared numeral suggests a question to resolve; it doesn't establish that those components operate together.

07 **Mira:** I have the register here already. Can we resolve the question through the records process instead of asking someone to go onto the roof?

08 **Owen:** Yes. Send the relevant [4] _____ entries and roof-plan reference. We're reconciling existing documents, not requesting physical access or a new equipment inspection.

09 **Mira:** Would it help if I changed AC-4 to CU-4 on the request now? At least that label would agree with the roof plan.

10 **Owen:** Please retain the original [5] _____ wording. Add the plan and register references alongside it; changing the label prematurely could narrow what the requester meant.

Conversation continues on the next page.

AC-4, CU-4, and AH-4 are not yet a pairing

CONVERSATION / CONTINUED

11 **Mira:** Right, a request about the whole system might become one about just the outdoor unit. We need the relationship before choosing the final reference.

12 **Owen:** Exactly. A [6] _____ and a component identifier can describe different levels of the equipment arrangement. The records need a supported link between them.

13 **Mira:** There may be model and serial fields in the register too. Those aren't interchangeable with the school's asset tags, are they?

14 **Owen:** No. A [7] _____ identifies a product design or version; a serial number distinguishes a manufactured unit. Keep each field with its recorded equipment item.

15 **Mira:** Another request mentions the meeting room. Can I borrow that location to make this one more useful, or would that introduce an unsupported connection?

16 **Owen:** Don't borrow it. The [8] _____ still needs confirmation for these references. A plausible location from a different request isn't evidence for this equipment relationship.

17 **Mira:** Please phrase the query without saying one document must be wrong. The labels may describe different things correctly, even if their connection is missing.

18 **Owen:** Ask for a confirmed [9] _____ between request AC-4, plan item CU-4, and register item AH-4, including the component relationship and the served area.

19 **Mira:** I'll send all three sources intact. That should let the record owner investigate the actual mapping instead of guessing which label we preferred.

20 **Owen:** Once we receive [10] _____, we can clarify the service reference. Until then, keep the separate labels and unresolved relationship visible in the request.

CHECK THE GAPS

1. record source / **2.** air handler / **3.** equipment pairing / **4.** equipment register / **5.** service request / **6.** system reference / **7.** model number / **8.** served area / **9.** cross-reference / **10.** confirmed mapping

Why these answers fit, page 119

F Check accuracy and scope

AC-4, CU-4, and AH-4 are not yet a pairing

Return to the case

A school service request calls the system AC-4. The roof plan labels an outdoor unit CU-4, while the indoor equipment register identifies air handler AH-4. Their relationship is not confirmed. Facilities manager Mira has the equipment register and discusses the request with service coordinator Owen. They need to preserve the three source references and seek confirmation through the records process. Neither person needs to access the roof or inspect equipment to clarify the existing documents.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Why is the numeral four insufficient?

A. AH always means the entire outdoor system
B. Similar numbering does not establish verified component relationships
C. Numbers can never be used in asset labels
D. All outdoor units must have different numerals

2. What should remain visible until clarified?

A. The source of each label and the unverified link
B. Only the shortest label
C. A presumed served area
D. An invented nameplate reading

Answers and explanations, page 120

Listen for the distinction

Switch roles for turns 11-20. Contrast system, component, model, serial, and served area without merging those fields.

Treat a familiar label as a starting point

Site shorthand may name a whole system while formal documents identify separate components. Similar numbering can suggest a question, but it is not proof of connection. Cite each label with its source so the next person can reconcile the records without silently merging them.

F Say it / transfer to a new case

Keep all three references

Complete the identification exchange using the three labels and the unresolved relationship. Use existing records only.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

AC-4 / AH-4 / unverified / CU-4

01 **Manager:** The request says [1] _____.

02 **Coordinator:** The roof plan identifies the outdoor unit as [2] _____.

03 **Manager:** The register identifies the air handler as [3] _____.

04 **Coordinator:** Their relationship remains [4] _____.

CHECK THE GAPS

1. AC-4 / **2.** CU-4 / **3.** AH-4 / **4.** unverified

Why these answers fit, page 121

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "There may be model and serial fields in the register too. Those aren't interchangeable with the school's asset tags, are they?" Owen: [missing response] Mira: "Another request mentions the meeting room. Can I borrow that location to make this one more useful, or would that introduce an unsupported connection?" Which response belongs in the gap?

A. Please retain the original service request wording. Add the plan and register references alongside it; changing the label prematurely could narrow what the requester meant.

B. Let's keep the record source beside each identifier. What does the roof plan call the outdoor equipment, and what does the indoor register say?

C. Don't borrow it. The served area still needs confirmation for these references. A plausible location from a different request isn't evidence for this equipment relationship.

D. No. A model number identifies a product design or version; a serial number distinguishes a manufactured unit. Keep each field with its recorded equipment item.

Answers and explanations, page 122

Review 1 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Clarifying comfort complaints and thermostat readings" and "Identifying equipment and linked components". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Target value used by a control system. Which term fits?

A. heat gain

B. setpoint

C. occupant feedback

D. target temperature

2. A colleague needs the term for this meaning: Equipment assembly located within the building or served space. Which term fits?

A. cross-reference

B. indoor unit

C. service request

D. equipment register

3. Rebuild the exchange from "Twenty-two is the target, not the finding". Asha: "I had written possible sensor failure to make the request more specific. But the information doesn't establish a failed sensor, does it?" Ben: [missing response] Asha: "Please include Celsius. Our overseas colleagues sometimes use Fahrenheit, and I'd rather the unit didn't disappear when the request is forwarded." Which response belongs in the gap?

A. It is. No equipment fault has been established. We should not name a failed sensor or another component simply to give the complaint a technical-sounding cause.

B. Yes. The missing temperature reading is a limit of this information. It does not establish the history of every measurement or the condition of the equipment.

C. Here is the attributed report: staff feel warm in the meeting room after 14:00; thermostat setpoint 22 degrees Celsius; no confirmed ambient reading supplied; cause unestablished.

D. I will pass on that record. It distinguishes the target temperature from the actual condition that still needs assessment, while keeping the staff's reported experience visible.

Review 1 / Meaning and context / continued

4. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "Please phrase the query without saying one document must be wrong. The labels may describe different things correctly, even if their connection is missing." Owen: [missing response] Mira: "I'll send all three sources intact. That should let the record owner investigate the actual mapping instead of guessing which label we preferred." Which response belongs in the gap?

A. Let's keep the record source beside each identifier. What does the roof plan call the outdoor equipment, and what does the indoor register say?

B. Yes. Send the relevant equipment register entries and roof-plan reference. We're reconciling existing documents, not requesting physical access or a new equipment inspection.

C. Then leave the equipment pairing unverified. The shared numeral suggests a question to resolve; it doesn't establish that those components operate together.

D. Ask for a confirmed cross-reference between request AC-4, plan item CU-4, and register item AH-4, including the component relationship and the served area.

Answers and explanations, page 153

A Read the situation

LESSON 03 / EXPLAINING AIRFLOW AND DISTRIBUTION REPORTS

Translate supply and return labels into plain language while preserving the difference between a comfort report and an airflow measurement.

> **Two ceiling openings, two functions**
>
> Client Tomas calls both ceiling openings air-conditioning outlets. The current drawing identifies S3 as supply air and R2 as return air. He reports discomfort below S3 but supplies no airflow measurements. Technician Leila explains the labels and records the location of the complaint. The drawing establishes the stated functions, not measured performance. No cause, balancing result, or adjustment is supplied, and the conversation must not prescribe changes to grilles or controls.

People: Tomas, Client; Leila, Technician.

Read the record

Choose one answer per question using the facts above.

1. Which opening does the drawing identify as supply?

A. Neither opening
B. S3
C. R2
D. Both openings

2. Where does Tomas report discomfort?

A. At an identified outdoor unit
B. Below S3
C. Below R2 only
D. In every room

3. What is missing?

A. The existence of two openings
B. The client's location description
C. Airflow measurements
D. The drawing labels

Answers and explanations, page 123

B Vocabulary in use

CORE TERMS

supply air
Air delivered into a served space through the distribution system.

identify supply air

return air
Air drawn from a space back toward the air-handling system.

identify return air

supply grille
Grille associated with delivery of air into the space.

locate the supply grille

return grille
Grille associated with air returning from the space.

identify the return grille

diffuser
Air-distribution device designed to spread delivered air.

identify the diffuser

register
Grille assembly that may include an airflow-control damper.

distinguish the register

duct
Passage used to convey air.

identify the duct

ductwork
Network of ducts and associated components.

review the ductwork

air distribution
Delivery and movement of air within a served area.

describe air distribution

airflow rate
Volume of air passing a point per unit time.

report the airflow rate

cubic feet per minute
Volume-flow unit commonly abbreviated CFM.

state cubic feet per minute

liters per second
Volume-flow unit used for air or liquid movement.

record liters per second

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

supply air SUPPLY AIR

identify supply air

Keep the meaning precise: Air delivered into a served space through the distribution system.

register REGISTER

distinguish the register

Keep the meaning precise: Grille assembly that may include an airflow-control damper.

B Precision and professional shorthand

EXTEND YOUR RANGE

air velocity

Speed of air movement at a stated position.

distinguish air velocity

air changes per hour

Hourly airflow divided by room volume, with the air source specified.

distinguish total and outdoor-air changes

draft

Air movement perceived as unwanted by an occupant.

describe a reported draft

occupied zone

Part of a room normally used by people.

identify the occupied zone

throw

Distance associated with a defined air jet under stated conditions.

review diffuser throw

spread

Lateral extent of a delivered air stream under defined conditions.

distinguish spread from throw

damper

Component used to regulate or control airflow in a system.

identify the damper

balancing

Technical adjustment and verification of distribution to meet specified requirements.

distinguish balancing from explanation

design airflow

Airflow value specified by the design.

compare design airflow

measured airflow

Airflow established through the relevant measurement.

record measured airflow

distribution report

Account describing air delivery or related comfort observations.

clarify the distribution report

performance finding

Supported conclusion about how a system functions.

distinguish a performance finding

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

liters per second LITERS PER SECOND

record liters per second

Keep the meaning precise: Volume-flow unit used for air or liquid movement.

throw THROW

review diffuser throw

Keep the meaning precise: Distance associated with a defined air jet under stated conditions.

C Phrases that move the work forward

Name the two functions

S3 is supply air; R2 is return air.

Explain return

Return air goes back toward the air-handling system.

Keep the complaint location

You report discomfort below S3.

Separate rate and speed

Airflow rate and air velocity are different quantities.

Avoid a balance claim

The labels do not establish that the system is balanced.

Clarify the everyday term

Calling both openings outlets hides their different functions.

Avoid prescribing a change

This explanation does not specify an adjustment.

Distinguish design from result

A design value is not automatically a measured result.

Explain supply

Supply air is delivered into the room.

Attribute the labels

Those are the functions shown on the current drawing.

Avoid converting experience into measurement

We do not have an airflow measurement for that location.

Name common units

Volume flow may be stated in cubic feet per minute or liters per second.

Keep cause open

The cause of the discomfort has not been established.

Retain the customer's words

I will record the discomfort without inventing a measured excess.

Ask for the existing reference

Please keep S3 attached to the reported location.

Close with a precise report

Discomfort below supply S3; airflow data not supplied; cause unresolved.

Precision in real exchanges

Purpose: Attribute the labels
Useful wording: Those are the functions shown on the current drawing.

Purpose: Keep the complaint location
Useful wording: You report discomfort below S3.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which explanation is accurate?

A. Supply means measured excess airflow

B. Supply delivers air into the room; return carries air back toward the system

C. Both openings must deliver air into the room

D. Return means outdoor exhaust in every system

2. What can the drawing labels establish here?

A. The cause of the discomfort

B. The correct damper adjustment

C. The stated functions of S3 and R2

D. A verified airflow rate

3. A colleague needs the term for this meaning: Grille associated with air returning from the space. Which term fits?

A. draft

B. air distribution

C. return grille

D. design airflow

D Choose the response that fits / continued

4. Rebuild the exchange from "Two ceiling openings, two functions". Tomas: "Then the complaint belongs below the supply opening, S3. Please keep that location rather than writing that the whole room's ventilation has failed." Leila: [missing response] Tomas: "I was about to say S3 delivers too much air. That's how it feels here, but I don't have a measured figure to attach." Which response belongs in the gap?

A. I'll retain it in the distribution report: discomfort below S3. We can identify the drawing label without claiming that it verifies how the installed system performs.

B. It means cubic feet per minute, a volume per unit time. It describes how much air volume moves, not the speed at one point by itself.

C. R2 is labeled return air. Supply delivers air into the space; return carries air back toward the air-handling system. Return isn't automatically an outdoor exhaust.

D. Correct. Air velocity is speed at a location, distinct from volume flow. Neither measurement is supplied in this report, although your location description is useful.

Answers and explanations, page 123

> **One useful distinction**
>
> S3 is labeled supply and R2 return on the current drawing. This tells the reader their stated functions. It does not establish actual flow rate, velocity, balance, or the cause of the discomfort.

E Conversation

Two ceiling openings, two functions

Two ceiling openings, two functions

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

supply air / Air velocity / performance finding / measured airflow / return air / occupied zone / damper / cubic feet per minute / balancing / distribution report

01 **Tomas:** Leila, the ceiling outlet above this seat is uncomfortable to sit under. It's S3 on the drawing. The other outlet is R2.

02 **Leila:** S3 is labeled [1] _____ on the drawing. R2 has a different function, so calling both outlets can hide a useful distinction.

03 **Tomas:** They look like parts of the same system. Does R2 also send air into the room, just from a different position?

04 **Leila:** R2 is labeled [2] _____. Supply delivers air into the space; return carries air back toward the air-handling system. Return isn't automatically an outdoor exhaust.

05 **Tomas:** Then the complaint belongs below the supply opening, S3. Please keep that location rather than writing that the whole room's ventilation has failed.

06 **Leila:** I'll retain it in the [3] _____: discomfort below S3. We can identify the drawing label without claiming that it verifies how the installed system performs.

07 **Tomas:** I was about to say S3 delivers too much air. That's how it feels here, but I don't have a measured figure to attach.

08 **Leila:** Keep it as your experience, then. A [4] _____ value needs measurement evidence. We shouldn't turn a comfort report into an unsupported volume-flow result.

09 **Tomas:** An earlier report used CFM. I assumed that was how fast the air moves. Does it actually describe something different?

10 **Leila:** It means [5] _____, a volume per unit time. It describes how much air volume moves, not the speed at one point by itself.

Conversation continues on the next page.

Two ceiling openings, two functions

CONVERSATION / CONTINUED

11 **Tomas:** So a feeling of air movement at my seat doesn't give us a CFM figure. And liters per second would be another volume-flow unit.

12 **Leila:** Correct. [6] _____ is speed at a location, distinct from volume flow. Neither measurement is supplied in this report, although your location description is useful.

13 **Tomas:** Since we know which opening is the return, can we at least say the system is balanced? The drawing appears to show a complete arrangement.

14 **Leila:** Identifying the openings doesn't establish [7] _____. That involves technical distribution work and verification, not simply checking that supply and return labels exist.

15 **Tomas:** Someone suggested closing a grille to stop the discomfort. I don't want the service request to present that suggestion as the approved remedy.

16 **Leila:** It won't. No [8] _____ adjustment or grille change is prescribed by these records. The appropriate assessment needs to establish the condition and any required response.

17 **Tomas:** Please include that it's the seat below S3 that concerns me. Saying air-conditioning problem leaves the person reading it with very little to locate.

18 **Leila:** Agreed. The [9] _____ location matters to the report. We'll preserve where you experience discomfort without naming an equipment fault that hasn't been established.

19 **Tomas:** Read me the short version, please. I'll use that wording when the facilities manager asks why the request was raised.

20 **Leila:** Discomfort reported below supply S3; R2 labeled return; no airflow measurements supplied. No [10] _____ or cause is established. That keeps the complaint precise without inventing results.

CHECK THE GAPS

1. supply air / **2.** return air / **3.** distribution report / **4.** measured airflow / **5.** cubic feet per minute / **6.** Air velocity / **7.** balancing / **8.** damper / **9.** occupied zone / **10.** performance finding

Why these answers fit, page 124

F Check accuracy and scope

Two ceiling openings, two functions

Return to the case

Client Tomas calls both ceiling openings air-conditioning outlets. The current drawing identifies S3 as supply air and R2 as return air. He reports discomfort below S3 but supplies no airflow measurements. Technician Leila explains the labels and records the location of the complaint. The drawing establishes the stated functions, not measured performance. No cause, balancing result, or adjustment is supplied, and the conversation must not prescribe changes to grilles or controls.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which report invents a measurement?

A. Tomas reports discomfort below S3
B. R2 is labeled return
C. No airflow data were supplied
D. S3 delivers too many liters per second

2. What is the proper scope of Leila's response?

A. Prescribe a grille adjustment
B. Declare the system balanced
C. Reject the complaint because S3 is supply
D. Explain labels and preserve the complaint for assessment

Answers and explanations, page 125

Listen for the distinction

Switch roles for turns 11-20. Distinguish CFM volume flow from air velocity and a comfort report from a measured result.

Translate the function, then return to the complaint

An everyday label can conceal an important functional difference. Explain the direction of airflow simply, then repeat the actual complaint. Technical vocabulary should improve the report, not overwhelm the customer or imply a diagnosis that the measurements do not support.

F Say it / transfer to a new case

Explain supply and return

Complete the client summary. Identify each function and keep the missing measurement visible.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

measurements / S3 / supply / return

01 **Technician:** S3 is labeled [1] _____ air.

02 **Client:** R2 is labeled [2] _____ air.

03 **Technician:** The discomfort is reported below [3] _____.

04 **Client:** No airflow [4] _____ have been supplied.

CHECK THE GAPS

1. supply / **2.** return / **3.** S3 / **4.** measurements

Why these answers fit, page 126

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Two ceiling openings, two functions". Tomas: "Since we know which opening is the return, can we at least say the system is balanced? The drawing appears to show a complete arrangement." Leila: [missing response] Tomas: "Someone suggested closing a grille to stop the discomfort. I don't want the service request to present that suggestion as the approved remedy." Which response belongs in the gap?

A. S3 is labeled supply air on the drawing. R2 has a different function, so calling both outlets can hide a useful distinction.

B. Keep it as your experience, then. A measured airflow value needs measurement evidence. We shouldn't turn a comfort report into an unsupported volume-flow result.

C. Identifying the openings doesn't establish balancing. That involves technical distribution work and verification, not simply checking that supply and return labels exist.

D. Agreed. The occupied zone location matters to the report. We'll preserve where you experience discomfort without naming an equipment fault that hasn't been established.

Answers and explanations, page 127

A Read the situation

LESSON 04 / CHECKING EQUIPMENT RATINGS AND REFRIGERANT REFERENCES

Compare like quantities, request missing equipment data, and distinguish refrigerant identification from approval to select or service a system.

Five kilowatts of what?

Client Jules compares two fictional proposals with estimator Farah. Both show 5 kilowatts, but proposal A labels the value cooling capacity while proposal B labels it electrical input. Jules calls them equal-size systems. Full data sheets and equipment pairings are not supplied, and no selection is approved. Farah must explain why the matching numbers are not enough, request complete specifications, and keep refrigerant references tied to the actual equipment without recommending handling or substitution.

People: Jules, Client; Farah, Estimator.

Read the record

Choose one answer per question using the facts above.

1. What does proposal A's 5 kilowatts describe?

A. Stored electrical energy
B. A refrigerant charge
C. Cooling capacity
D. Electrical input

2. What does proposal B's 5 kilowatts describe?

A. Confirmed equal cooling capacity
B. A measured room temperature
C. A maintenance interval
D. Electrical input

3. What conclusion is supported?

A. Both have identical efficiency
B. Proposal B is approved
C. The numbers alone cannot establish equivalent systems
D. Both systems have identical capacity

Answers and explanations, page 128

B Vocabulary in use

CORE TERMS

cooling capacity
Rate at which a system can remove heat under stated conditions.

compare cooling capacity

electrical input
Electrical power required by the equipment under stated conditions.

identify electrical input

kilowatt
Unit of power equal to one thousand watts.

state the rating in kilowatts

kilowatt-hour
Unit of energy representing one kilowatt over one hour.

distinguish a kilowatt-hour

rating condition
Defined condition under which a performance value is stated.

check the rating conditions

data sheet
Manufacturer document listing relevant product characteristics.

obtain the complete data sheet

equipment combination
Particular set of components proposed to work together.

verify the equipment combination

matched system
Component arrangement confirmed as suitable together by the relevant specifications.

identify the matched system

nominal capacity
Stated classification or approximate rating rather than every operating output.

distinguish nominal capacity

operating condition
Relevant environmental or system state during operation.

state the operating conditions

efficiency
Relationship between useful output and required input under defined conditions.

compare efficiency on a consistent basis

coefficient of performance
Ratio of useful heating or cooling output to input power on a defined basis.

interpret the coefficient of performance

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

cooling capacity COOLING CAPACITY

compare cooling capacity

Keep the meaning precise: Rate at which a system can remove heat under stated conditions.

data sheet DATA SHEET

obtain the complete data sheet

Keep the meaning precise: Manufacturer document listing relevant product characteristics.

B Precision and professional shorthand

EXTEND YOUR RANGE

refrigerant

Working fluid used to transfer heat in a refrigeration system.

identify the specified refrigerant

refrigerant charge

Quantity of refrigerant specified or present in a system.

distinguish refrigerant charge

compatibility

Suitability of materials or components for use together.

verify compatibility

superheat

Vapor temperature above its saturation reference at the same pressure; use dew for the stated blend.

report superheat in kelvin

like-for-like comparison

Comparison of matching quantities on a consistent basis.

make a like-for-like comparison

refrigerant designation

Identifier naming a particular refrigerant or blend.

preserve the refrigerant designation

blend

Refrigerant made from more than one component substance.

identify a refrigerant blend

manufacturer specification

Product requirement or attribute stated by its manufacturer.

consult the manufacturer specification

selection basis

Information and criteria supporting an equipment choice.

establish the selection basis

sensible heat ratio

Sensible cooling capacity divided by total cooling capacity on the same basis.

calculate the sensible heat ratio

B Precision and professional shorthand / continued

subcooling

Liquid temperature below its saturation reference at the same pressure; use bubble for the stated blend.

distinguish subcooling from superheat

substitution review

Evaluation of a proposed replacement for a specified item.

request a substitution review

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

coefficient of performance COEFFICIENT OF PERFORMANCE

interpret the coefficient of performance

Keep the meaning precise: Ratio of useful heating or cooling output to input power on a defined basis.

compatibility COMPATIBILITY

verify compatibility

Keep the meaning precise: Suitability of materials or components for use together.

C Phrases that move the work forward

Ask the key question
Five kilowatts of which quantity?

Read proposal A
A lists 5 kilowatts of cooling capacity.

Read proposal B
B lists 5 kilowatts of electrical input.

Name the distinction
Cooling output and electrical input are different.

Reject the false equivalence
Matching numbers do not establish equal-size systems here.

Ask for complete data
We need the full data sheets for both proposals.

Include the combination
The proposed component pairings also need to be identified.

Keep conditions attached
Compare ratings under the relevant stated conditions.

Avoid inventing efficiency
We cannot calculate a supported efficiency comparison from these two entries.

Distinguish power and energy
Kilowatts and kilowatt-hours are different units.

Keep refrigerant information specific
Use the refrigerant designation from the actual equipment specification.

Avoid treating a name as approval
A refrigerant reference does not approve a substitution.

Keep charge separate
Refrigerant type and refrigerant quantity are different information.

Name the review task
We need a like-for-like comparison.

State the selection limit
No equipment selection has been approved.

Close with the request
Please supply complete ratings and equipment combinations for review.

Precision in real exchanges

Purpose: Name the distinction
Useful wording: Cooling output and electrical input are different.

Purpose: Reject the false equivalence
Useful wording: Matching numbers do not establish equal-size systems here.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which comparison is valid from the supplied entries?

A. Both values prove identical energy consumption
B. B must be more efficient than A
C. A's value is cooling capacity; B's is electrical input
D. Both values prove equal cooling output

2. What additional information is needed?

A. Confirmation that both headlines still say five kilowatts
B. A shared refrigerant name without the component information
C. Full data sheets, rating conditions, and component pairings
D. A ratio of A's output to B's input

3. A colleague needs the term for this meaning: Unit of energy representing one kilowatt over one hour. Which term fits?

A. like-for-like comparison
B. refrigerant charge
C. kilowatt-hour
D. nominal capacity

D Choose the response that fits / continued

4. Rebuild the exchange from "Five kilowatts of what?". Jules: "Could I divide A's five by B's five and report a performance ratio of one? They're both power figures, after all." Farah: [missing response] Jules: "Then asking each supplier to confirm five kilowatts again wouldn't fix it. We need both quantity labels and the missing values for each proposal." Which response belongs in the gap?

A. Electrical input is the power required by the equipment. Cooling capacity is heat-removal output under stated conditions, so these entries don't establish equal output.

B. Not for an efficiency comparison. You would be combining one system's output with another system's input, rather than corresponding quantities for each proposed arrangement.

C. Yes, request each full data sheet, with the proposed components and rating conditions. A short headline doesn't establish performance across every operating situation.

D. That improves the selection basis. It doesn't approve either option yet, but it gives the technical review the information needed to make a meaningful comparison.

Answers and explanations, page 128

> **One useful distinction**
>
> Both proposals state 5 kilowatts, but A describes cooling output and B electrical input. The unit can express either quantity. It does not follow that the systems have equal cooling capacities or equal efficiencies.

E Conversation

Five kilowatts of what?

Five kilowatts of what?

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

data sheet / Refrigerant charge / substitution review / efficiency / selection basis / refrigerant designation / Electrical input / like-for-like comparison / equipment combination / cooling capacity

01 **Jules:** Farah, both proposals show five kilowatts. I put equal-size systems in my comparison sheet and moved straight to the two prices.

02 **Farah:** Let's check five kilowatts of what. Proposal A labels it [1] _____. B uses a different quantity label even though its number and unit look the same.

03 **Jules:** B says electrical input. I treated kilowatts as though they always told us how much heat the system could remove from the room.

04 **Farah:** [2] _____ is the power required by the equipment. Cooling capacity is heat-removal output under stated conditions, so these entries don't establish equal output.

05 **Jules:** Could I divide A's five by B's five and report a performance ratio of one? They're both power figures, after all.

06 **Farah:** Not for an [3] _____ comparison. You would be combining one system's output with another system's input, rather than corresponding quantities for each proposed arrangement.

E Conversation / continued

07 **Jules:** Then asking each supplier to confirm five kilowatts again wouldn't fix it. We need both quantity labels and the missing values for each proposal.

08 **Farah:** Yes, request each full [4] _____, with the proposed components and rating conditions. A short headline doesn't establish performance across every operating situation.

09 **Jules:** I haven't received the indoor and outdoor pairings. Is a brochure for the outdoor unit enough if the installer chooses the indoor one later?

10 **Farah:** No. The [5] _____ matters to the stated performance. We need to know which components the proposal actually pairs, not assume any indoor unit gives the same result.

Conversation continues on the next page.

Five kilowatts of what?

CONVERSATION / CONTINUED

11 **Jules:** One summary includes a refrigerant name. Could I use that as evidence that the two proposals are interchangeable replacements?

12 **Farah:** A [6] _____ identifies a fluid or blend, not an equivalent system. It must be checked against the actual equipment specification and doesn't settle all compatibility requirements.

13 **Jules:** I don't want the supplier to read our question as permission to use another fluid. We're asking for information, not changing the specified equipment.

14 **Farah:** Then keep any [7] _____ separate and explicitly pending. A familiar refrigerant name doesn't bypass that process or authorize refrigerant handling.

15 **Jules:** The sheet also has a refrigerant amount. Is that a different statement of capacity, or is it simply another product detail?

16 **Farah:** [8] _____ is the quantity of refrigerant specified or present. Keep the type and amount separate from cooling capacity and electrical input.

17 **Jules:** I see four different things now: fluid identity, fluid quantity, output power, and input power. The matching fives only concern two of those.

18 **Farah:** Right. We need a [9] _____ using corresponding ratings and conditions. None of the missing values should be filled with the figure from the other proposal.

19 **Jules:** I'll remove equal-size and ask for complete data sheets, actual pairings, and both output and input values with their conditions.

20 **Farah:** That improves the [10] _____. It doesn't approve either option yet, but it gives the technical review the information needed to make a meaningful comparison.

CHECK THE GAPS

1. cooling capacity / **2.** Electrical input / **3.** efficiency / **4.** data sheet / **5.** equipment combination / **6.** refrigerant designation / **7.** substitution review / **8.** Refrigerant charge / **9.** like-for-like comparison / **10.** selection basis

Why these answers fit, page 129

F Check accuracy and scope

Five kilowatts of what?

Return to the case

Client Jules compares two fictional proposals with estimator Farah. Both show 5 kilowatts, but proposal A labels the value cooling capacity while proposal B labels it electrical input. Jules calls them equal-size systems. Full data sheets and equipment pairings are not supplied, and no selection is approved. Farah must explain why the matching numbers are not enough, request complete specifications, and keep refrigerant references tied to the actual equipment without recommending handling or substitution.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which conclusion exceeds the evidence?

A. The systems have equal efficiency
B. A is labeled cooling capacity
C. B is labeled electrical input
D. No selection is approved

2. What does a refrigerant designation provide?

A. A complete compatibility assessment
B. Permission for untrained servicing
C. Identification to be checked against the actual equipment specification
D. Automatic approval to substitute fluids

Answers and explanations, page 131

Listen for the distinction

Switch roles for turns 11-20. Keep refrigerant designation and charge separate, and make the missing pairing explicit.

Ask what a familiar number measures

Customers often compare the largest printed number in a proposal. Name the quantity before comparing the value. Explain what information is missing and why it matters, without turning a partial specification into a product recommendation or an efficiency claim.

F Say it / transfer to a new case

Keep input and output apart

Complete the proposal comparison. Preserve the different quantity labels and the missing approval.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

sheets / input / approved / capacity

01 **Client:** A gives 5 kilowatts of cooling [1] _____.

02 **Estimator:** B gives 5 kilowatts of electrical [2] _____.

03 **Client:** We need complete data [3] _____ and pairings.

04 **Estimator:** No selection is yet [4] _____.

CHECK THE GAPS

1. capacity / **2.** input / **3.** sheets / **4.** approved

Why these answers fit, page 131

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Five kilowatts of what?". Jules: "I don't want the supplier to read our question as permission to use another fluid. We're asking for information, not changing the specified equipment." Farah: [missing response] Jules: "The sheet also has a refrigerant amount. Is that a different statement of capacity, or is it simply another product detail?" Which response belongs in the gap?

A. Yes, request each full data sheet, with the proposed components and rating conditions. A short headline doesn't establish performance across every operating situation.

B. Refrigerant charge is the quantity of refrigerant specified or present. Keep the type and amount separate from cooling capacity and electrical input.

C. Then keep any substitution review separate and explicitly pending. A familiar refrigerant name doesn't bypass that process or authorize refrigerant handling.

D. Electrical input is the power required by the equipment. Cooling capacity is heat-removal output under stated conditions, so these entries don't establish equal output.

Answers and explanations, page 132

Review 2 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Explaining airflow and distribution reports" and "Checking equipment ratings and refrigerant references". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Grille associated with delivery of air into the space. Which term fits?

A. supply grille
B. balancing
C. ductwork
D. air changes per hour

2. A colleague needs the term for this meaning: Component arrangement confirmed as suitable together by the relevant specifications. Which term fits?

A. matched system
B. superheat
C. cooling capacity
D. refrigerant

3. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "Would it help if I changed AC-4 to CU-4 on the request now? At least that label would agree with the roof plan." Owen: [missing response] Mira: "Right, a request about the whole system might become one about just the outdoor unit. We need the relationship before choosing the final reference." Which response belongs in the gap?

A. Please retain the original service request wording. Add the plan and register references alongside it; changing the label prematurely could narrow what the requester meant.
B. Don't borrow it. The served area still needs confirmation for these references. A plausible location from a different request isn't evidence for this equipment relationship.
C. Let's keep the record source beside each identifier. What does the roof plan call the outdoor equipment, and what does the indoor register say?
D. No. A model number identifies a product design or version; a serial number distinguishes a manufactured unit. Keep each field with its recorded equipment item.

Review 2 / Meaning and context / continued

4. Rebuild the exchange from "Five kilowatts of what?". Jules: "I see four different things now: fluid identity, fluid quantity, output power, and input power. The matching fives only concern two of those." Farah: [missing response] Jules: "I'll remove equal-size and ask for complete data sheets, actual pairings, and both output and input values with their conditions." Which response belongs in the gap?

A. A refrigerant designation identifies a fluid or blend, not an equivalent system. It must be checked against the actual equipment specification and doesn't settle all compatibility requirements.

B. Let's check five kilowatts of what. Proposal A labels it cooling capacity. B uses a different quantity label even though its number and unit look the same.

C. Right. We need a like-for-like comparison using corresponding ratings and conditions. None of the missing values should be filled with the figure from the other proposal.

D. That improves the selection basis. It doesn't approve either option yet, but it gives the technical review the information needed to make a meaningful comparison.

Answers and explanations, page 156

A Read the situation

LESSON 05 / REPORTING MOISTURE, SYMPTOMS, AND SERVICE FINDINGS

Reconcile a historical report with a later observation and avoid treating an earlier maintenance concern as a proven cause.

Yesterday wet, today dry, source still unknown

Client Elena reported water near air handler AH-6 at 15:00 yesterday. Today's visit note states that the nearby floor is dry and the source was not identified. An earlier service log mentions a drain-pan concern but does not establish a link to the reported water. Coordinator Kai must preserve the chronology, distinguish each source, and summarize the unresolved issue. The exercise supplies records only; no equipment inspection or diagnostic procedure is requested.

People: Elena, Client; Kai, Service coordinator.

Read the record

Choose one answer per question using the facts above.

1. What did Elena report yesterday?

A. A verified drain-pan failure
B. A completed repair
C. A dry floor throughout the day
D. Water near AH-6 at 15:00

2. What does today's visit note establish?

A. The earlier report was false
B. The drain pan caused the water
C. Every component was repaired
D. The nearby floor was dry and the source was not identified

3. What does the earlier drain-pan entry prove?

A. Only that a concern was recorded, not a causal link
B. That the pan caused yesterday's water
C. That the concern has been repaired
D. That the current system is fault-free

Answers and explanations, page 133

B Vocabulary in use

CORE TERMS

condensate
Liquid formed when water vapor condenses.

identify a condensate reference

condensate line
Pipe or tube associated with carrying condensate away.

identify the condensate line

service log
Chronological record of service-related information.

consult the service log

historical entry
Record created before the current event.

distinguish a historical entry

reported water
Water described in an account before its cause is established.

locate the reported water

source identification
Establishing where the reported water originated.

keep source identification open

drain pan
Component intended to collect relevant drainage or condensate in equipment.

record a drain-pan concern

moisture report
Account describing water or dampness at a location.

retain the moisture report

visit note
Record of observations or actions during an attendance.

quote the visit note

chronology
Ordered sequence of events and observations.

preserve the chronology

nearby floor
Floor surface close to the referenced equipment.

describe the nearby floor

causal link
Supported connection between a condition and its cause.

establish a causal link

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

condensate CONDENSATE

identify a condensate reference

Keep the meaning precise: Liquid formed when water vapor condenses.

visit note VISIT NOTE

quote the visit note

Keep the meaning precise: Record of observations or actions during an attendance.

B Precision and professional shorthand

EXTEND YOUR RANGE

unresolved concern

Issue that has not been confirmed as corrected or closed.

retain an unresolved concern

intermittent symptom

Condition that may appear at some times and not others.

describe an intermittent symptom

drainage path

Route intended for the relevant liquid to travel.

identify the drainage path

leakage

Unintended escape of a fluid.

avoid assuming leakage source

finding

Observation or conclusion established through the relevant assessment.

state the actual finding

evidence boundary

Limit of what the information supports.

preserve the evidence boundary

dry observation

Record that the visible surface was dry at the stated time.

qualify the dry observation

recurrence history

Record of repeated appearances of a condition.

verify recurrence history

overflow

Liquid exceeding the capacity or boundary of its intended containment.

distinguish a reported overflow

condensation

Formation of liquid from vapor under relevant conditions.

distinguish condensation from a diagnosis

repair status

Whether corrective work is proposed, underway, or completed.

verify repair status

record reconciliation

Bringing separate accounts together without erasing their distinctions.

complete record reconciliation

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

causal link CAUSAL LINK

establish a causal link

Keep the meaning precise: Supported connection between a condition and its cause.

drainage path DRAINAGE PATH

identify the drainage path

Keep the meaning precise: Route intended for the relevant liquid to travel.

C Phrases that move the work forward

State yesterday's account

The client reported water near AH-6 at 15:00 yesterday.

State today's observation

The visit note records the nearby floor as dry today.

Keep both times

Those accounts refer to different observations at different times.

Name the unresolved issue

The source was not identified.

Attribute the history

An earlier service-log entry mentions a drain-pan concern.

Avoid converting history into cause

That entry does not establish a link to yesterday's water.

Distinguish near from from

Near AH-6 gives a location, not a proven origin.

Avoid changing concern to failure

The record says concern, not confirmed failure.

Keep repair separate

No completed repair is established by the dry-floor note.

Preserve the client report

The later observation does not cancel your account.

Use the exact equipment reference

Please retain AH-6 in the summary.

Avoid an invented recurrence pattern

We have not established how often this happens.

Name the three sources

Client report, visit note, and earlier log remain distinct.

Limit the finding

We can report the surface condition without diagnosing the source.

Give a balanced summary

Water reported yesterday; floor dry today; cause unresolved.

Close with accurate status

The source question remains open for the relevant assessment.

Precision in real exchanges

Purpose: Name the unresolved issue
Useful wording: The source was not identified.

Purpose: Attribute the history
Useful wording: An earlier service-log entry mentions a drain-pan concern.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which summary preserves chronology?

A. Water from AH-6 yesterday at 15:00; floor dry today

B. Water seen by the service team today; earlier pan concern

C. Drain-pan failure yesterday; repair inferred from today's dry floor

D. Water near AH-6 yesterday at 15:00; floor dry today; source unidentified

2. Which wording improperly asserts origin?

A. The nearby floor was dry today

B. The source was not identified

C. Water came from AH-6

D. Water was reported near AH-6

3. A colleague needs the term for this meaning: Account describing water or dampness at a location. Which term fits?

A. moisture report

B. finding

C. reported water

D. intermittent symptom

D Choose the response that fits / continued

4. Rebuild the exchange from "Yesterday wet, today dry, source still unknown". Elena: "The earlier service log mentions a drain-pan concern. Someone suggested putting that as the cause so the record has an explanation." Kai: [missing response] Elena: "I only know the water was near the air handler. Writing that it came from AH-6 would be more specific than what I observed." Which response belongs in the gap?

A. We should retain the drain-pan concern as history, but the entry does not establish a connection to yesterday's report. It is not enough to name the pan as the proven source.

B. Client reported water near AH-6 at 15:00 yesterday. Today's floor was dry; source unidentified. Earlier log notes a drain-pan concern, with no established causal link to the water.

C. It would change it. The historical entry says concern, not confirmed failure. We need to preserve that distinction even if the topic appears relevant to the current question.

D. I will use that record reconciliation in the handoff. The next assessment can address the source question with the original accounts intact, rather than starting from an assumed diagnosis.

Answers and explanations, page 133

One useful distinction

Yesterday at 15:00, Elena reported water near AH-6. Today, the floor is recorded as dry. Those observations concern different times; the later note does not disprove the earlier report or establish a repair.

E Conversation

Yesterday wet, today dry, source still unknown

Yesterday wet, today dry, source still unknown

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

chronology / historical entry / causal link / drain-pan concern / evidence boundary / visit note / source identification / repair status / record reconciliation / recurrence history

01 **Elena:** Kai, yesterday I reported water near AH-6 at three. Today's note says dry floor. I don't want my report dropped because the visit happened later.

02 **Kai:** We can preserve the [1] _____. Your account and today's note refer to different times, so a dry floor now does not cancel the water you reported yesterday.

03 **Elena:** Please keep yesterday and fifteen hundred attached to my account. Otherwise it could sound as though the technician saw water on today's visit.

04 **Kai:** I will. The [2] _____ records today's nearby floor as dry and says the source was not identified. That is a separate observation with its own limits.

05 **Elena:** The earlier service log mentions a drain-pan concern. Someone suggested putting that as the cause so the record has an explanation.

06 **Kai:** We should retain the [3] _____ as history, but the entry does not establish a connection to yesterday's report. It is not enough to name the pan as the proven source.

07 **Elena:** I only know the water was near the air handler. Writing that it came from AH-6 would be more specific than what I observed.

08 **Kai:** Exactly. That is an [4] _____. Changing near AH-6 to from AH-6 would turn a location description into an unsupported statement about origin.

09 **Elena:** The old wording is concern, not failure. Could you retain that distinction rather than turning a historical note into a confirmed broken part?

10 **Kai:** It would change it. The [5] _____ says concern, not confirmed failure. We need to preserve that distinction even if the topic appears relevant to the current question.

Conversation continues on the next page.

Yesterday wet, today dry, source still unknown

CONVERSATION / CONTINUED

11 **Elena:** Then we'll have an unresolved source, but an accurate record. I don't want a neat explanation that neither the client report nor visit note establishes.

12 **Kai:** Then the [6] _____ stays open. We can report the facts and the earlier concern without claiming that the relationship between them has been verified.

13 **Elena:** Does dry today imply someone repaired it overnight? I haven't supplied any record of work between my report and the visit.

14 **Kai:** No. The [7] _____ cannot be inferred from the dry-floor observation. Nothing in these records establishes that a repair was completed between your report and the visit.

15 **Elena:** One older concern also doesn't establish a pattern of repeated water events. We shouldn't attach a frequency to the problem without the history.

16 **Kai:** Correct. A [8] _____ would need supporting information. We should not invent a frequency or treat every earlier note as another occurrence of the same water event.

17 **Elena:** Please combine the three records without losing their sources: my account, today's visit note, and the earlier drain-pan entry.

18 **Kai:** Client reported water near AH-6 at 15:00 yesterday. Today's floor was dry; source unidentified. Earlier log notes a drain-pan concern, with no established [9] _____ to the water.

19 **Elena:** That's the chronology I need. It neither dismisses yesterday's observation nor treats today's dry floor as proof that the source has been fixed.

20 **Kai:** I will use that [10] _____ in the handoff. The next assessment can address the source question with the original accounts intact, rather than starting from an assumed diagnosis.

CHECK THE GAPS

1. chronology / **2.** visit note / **3.** drain-pan concern / **4.** evidence boundary / **5.** historical entry / **6.** source identification / **7.** repair status / **8.** recurrence history / **9.** causal link / **10.** record reconciliation

Why these answers fit, page 134

F Check accuracy and scope

Yesterday wet, today dry, source still unknown

Return to the case

Client Elena reported water near air handler AH-6 at 15:00 yesterday. Today's visit note states that the nearby floor is dry and the source was not identified. An earlier service log mentions a drain-pan concern but does not establish a link to the reported water. Coordinator Kai must preserve the chronology, distinguish each source, and summarize the unresolved issue. The exercise supplies records only; no equipment inspection or diagnostic procedure is requested.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. How should the drain-pan entry be handled?

A. Delete it because the floor is dry
B. Relabel it as a completed repair
C. Retain it as history with no established causal link
D. Treat it as the final diagnosis

2. What can the dry-floor note establish about repair?

A. A drainage component was replaced
B. No completed repair by itself
C. Every possible leak was fixed
D. The earlier concern was resolved

Answers and explanations, page 136

Listen for the distinction

Switch roles for turns 11-20. Keep near distinct from from, and dry distinct from repaired.

Keep two times visible in one account

A later observation need not contradict an earlier report. Use time markers and source attribution rather than choosing one account and deleting the other. A prior concern can remain relevant history without becoming the confirmed cause of a new symptom.

F Say it / transfer to a new case

Preserve the three records

Complete the service summary. Keep yesterday, today, and the earlier concern distinct.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

15:00 / concern / dry / identified

01 **Client:** Water was reported near AH-6 at [1] _____ yesterday.

02 **Coordinator:** Today the nearby floor was recorded as [2] _____.

03 **Client:** The earlier log mentions a drain-pan [3] _____.

04 **Coordinator:** The source was not [4] _____.

CHECK THE GAPS

1. 15:00 / **2.** dry / **3.** concern / **4.** identified

Why these answers fit, page 136

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Yesterday wet, today dry, source still unknown". Elena: "Does dry today imply someone repaired it overnight? I haven't supplied any record of work between my report and the visit." Kai: [missing response] Elena: "One older concern also doesn't establish a pattern of repeated water events. We shouldn't attach a frequency to the problem without the history." Which response belongs in the gap?

A. I will. The visit note records today's nearby floor as dry and says the source was not identified. That is a separate observation with its own limits.

B. Correct. A recurrence history would need supporting information. We should not invent a frequency or treat every earlier note as another occurrence of the same water event.

C. We can preserve the chronology. Your account and today's note refer to different times, so a dry floor now does not cancel the water you reported yesterday.

D. No. The repair status cannot be inferred from the dry-floor observation. Nothing in these records establishes that a repair was completed between your report and the visit.

Answers and explanations, page 137

A Read the situation

LESSON 06 / DISCUSSING REFRIGERATION ALARMS AND TEMPERATURE RECORDS

Report an alarm chronology accurately, distinguish cabinet and product information, and route a food-safety question to the responsible lead.

An alarm, a display value, and missing product data

Cafe manager Noah discusses a refrigeration record with technician Imani. A cabinet display recorded 8 degrees Celsius at 06:10. Product temperatures are not supplied. An alarm-history entry and a scheduled-defrost entry have the same time, but event cause and duration are unconfirmed. Noah has already alerted the responsible food-safety lead under the site procedures; this discussion occurs alongside that escalation and does not delay protective action. He asks how the records relate to the food-safety question. The display value and coincident entries do not authorize food release or establish a harmless defrost event.

People: Noah, Cafe manager; Imani, Refrigeration technician.

Read the record

Choose one answer per question using the facts above.

1. What was recorded at 06:10?

A. A confirmed harmless alarm cause
B. The end of a measured exposure period
C. A cabinet display value of 8 degrees Celsius
D. Verified temperatures for every food item

2. What do the matching timestamps establish?

A. Every food item remained safe
B. The two entries share a recorded time
C. Defrost definitely caused the alarm
D. The alarm lasted one minute

3. Who must decide the food-safety question?

A. The responsible food-safety lead under site procedures
B. The display value alone
C. The technician by assuming defrost
D. The customer waiting at the counter

Answers and explanations, page 138

B Vocabulary in use

CORE TERMS

refrigerated cabinet
Enclosed equipment used to hold products under cooled conditions.

identify the refrigerated cabinet

cabinet display
Interface showing a value or status from refrigeration equipment.

quote the cabinet display

product temperature
Temperature of the actual stored product.

distinguish product temperature

display value
Number shown by the equipment interface.

record the display value

alarm history
Log of recorded equipment alarm events.

review the alarm history

timestamp
Recorded date or time associated with an entry.

preserve the timestamp

defrost
Process intended to remove frost from relevant equipment surfaces.

identify the defrost entry

scheduled defrost
Defrost event listed in the control schedule or record.

distinguish scheduled defrost

event duration
Length of time an event continued.

establish event duration

coincident entries
Records appearing at the same stated time.

compare coincident entries

causation
Relationship in which one event produces another.

avoid assuming causation

temperature excursion
Departure from a defined temperature range or limit.

assess a recorded temperature excursion

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

refrigerated cabinet REFRIGERATED CABINET

identify the refrigerated cabinet

Keep the meaning precise: Enclosed equipment used to hold products under cooled conditions.

timestamp TIMESTAMP

preserve the timestamp

Keep the meaning precise: Recorded date or time associated with an entry.

B Precision and professional shorthand

EXTEND YOUR RANGE

exposure history

Record of conditions experienced by the product over time.

establish exposure history

data gap

Missing information needed for an assessment.

identify the data gap

food-safety lead

Person assigned responsibility for the relevant food-safety decision.

notify the food-safety lead

site procedure

Established process applicable at the premises.

follow the site procedure

release decision

Authorization to make a product available for its intended use.

refer the release decision

product disposition

Decision about how affected products will be handled.

determine product disposition

alarm acknowledgment

Record that an alarm has been noticed or accepted by an operator.

distinguish alarm acknowledgment

alarm clearance

Change in alarm status that does not alone establish product safety.

distinguish alarm clearance

trend record

Series of values showing change over time.

review the trend record

sensor location

Position associated with a particular measurement device.

identify the sensor location

technical cause

Established equipment or process explanation for an event.

keep the technical cause open

escalation summary

Concise information passed to the responsible decision-maker.

give an escalation summary

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

temperature excursion TEMPERATURE EXCURSION

assess a recorded temperature excursion

Keep the meaning precise: Departure from a defined temperature range or limit.

release decision RELEASE DECISION

refer the release decision

Keep the meaning precise: Authorization to make a product available for its intended use.

C Phrases that move the work forward

Quote the actual record

The cabinet display recorded 8 degrees Celsius at 06:10.

Keep the record type clear

That is a display value, not a verified food-temperature record.

Avoid a causal shortcut

The matching time does not establish that defrost caused the alarm.

Reject premature reassurance

I cannot declare the food safe from these records.

Keep technical status open

The cause of the event still needs confirmation.

Preserve the data gap

The product exposure history is not established.

Keep disposition separate

Product disposition is separate from explaining the equipment entry.

Read back the essentials

8 degrees at 06:10; product data absent; cause and duration unconfirmed.

Name the missing information

Product temperatures have not been supplied.

Identify the coincident entries

The alarm and scheduled-defrost entries share a timestamp.

Preserve duration uncertainty

The event duration is unconfirmed.

Name the decision route

The responsible food-safety lead must decide under the site procedures.

Avoid equating clearance with safety

Alarm clearance would not itself release the food.

Escalate accurately

Please pass the display record and the missing data to the lead.

Avoid a guessed threshold

We should not invent an allowable exposure period.

Close without releasing product

This technical discussion does not authorize food release.

Precision in real exchanges

Purpose: Identify the coincident entries
Useful wording: The alarm and scheduled-defrost entries share a timestamp.

Purpose: Avoid a causal shortcut
Useful wording: The matching time does not establish that defrost caused the alarm.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which summary belongs in the escalation?

A. Every product was measured at 8 degrees

B. Alarm lasted only until 06:11

C. Display 8 degrees Celsius at 06:10; product data absent; cause and duration unconfirmed

D. Food definitely safe because defrost was scheduled

2. Why is defrost caused it unsupported?

A. A scheduled entry proves the entire cycle completed normally

B. A timestamp establishes both start and finish

C. A defrost entry establishes every product's exposure history

D. The shared timestamp does not establish causation

3. A colleague needs the term for this meaning: Number shown by the equipment interface. Which term fits?

A. event duration

B. display value

C. food-safety lead

D. trend record

D Choose the response that fits / continued

4. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "The alarm history and the scheduled-defrost entry both say 06:10. Could I describe the alarm as caused by defrost?" Imani: [missing response] Noah: "The owner asked how long it lasted. There's only that one recorded time in this extract, with no confirmed ending time." Which response belongs in the gap?

A. It contains a scheduled defrost entry at the same recorded time as the alarm. That association needs assessment; it isn't yet a confirmed causal explanation.

B. Then product temperature is missing. Keep that gap explicit; don't copy the display number into a field that would imply the actual food was measured.

C. That's a useful escalation summary. Send any relevant updates through the site process as they become available, keeping the product question separate from the equipment diagnosis.

D. The shared timestamp only tells us the entries coincide. It doesn't establish the cause, prove the event was harmless, or determine what happened to the products.

Answers and explanations, page 138

> **One useful distinction**
>
> The record gives a display value of 8 degrees Celsius at 06:10. Product temperatures are absent. No safe holding period or release decision can be inferred from this isolated display value and timestamp.

E Conversation

An alarm, a display value, and missing product data

An alarm, a display value, and missing product data

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

timestamp / Product disposition / event duration / Alarm clearance / escalation summary / exposure history / food-safety lead / product temperature / scheduled defrost / display value

01 **Noah:** Imani, we've alerted our food-safety lead and aren't treating this call as clearance to serve the affected food. Can you help explain the cabinet record?

02 **Imani:** Yes. The [1] _____ is eight degrees Celsius at 06:10. It describes the equipment record, not a verified temperature for every stored item.

03 **Noah:** That's all I've been given for temperature: the cabinet display. There are no food readings in the information I'm sending to the lead.

04 **Imani:** Then [2] _____ is missing. Keep that gap explicit; don't copy the display number into a field that would imply the actual food was measured.

05 **Noah:** The alarm history and the scheduled-defrost entry both say 06:10. Could I describe the alarm as caused by defrost?

06 **Imani:** The shared [3] _____ only tells us the entries coincide. It doesn't establish the cause, prove the event was harmless, or determine what happened to the products.

E Conversation / continued

07 **Noah:** The owner asked how long it lasted. There's only that one recorded time in this extract, with no confirmed ending time.

08 **Imani:** Then [4] _____ is unconfirmed. A timestamp doesn't mean a one-minute event, and we shouldn't invent an interval to make the report look complete.

09 **Noah:** I'll keep the lead updated with those gaps. Our opening time is close, but the technical call isn't a substitute for the product decision.

10 **Imani:** Correct. The [5] _____ must decide under your site procedures. Follow that process while the equipment information is clarified; don't wait for a reassuring guess from the display.

Conversation continues on the next page.

An alarm, a display value, and missing product data

CONVERSATION / CONTINUED

11 **Noah:** Please distinguish the scheduled item from the actual explanation. I want the handoff to say what the log contains without implying we've diagnosed the event.

12 **Imani:** It contains a [6] _____ entry at the same recorded time as the alarm. That association needs assessment; it isn't yet a confirmed causal explanation.

13 **Noah:** Suppose a later update says the alarm has disappeared. Would that alone answer whether the affected stock can be used?

14 **Imani:** No. [7] _____ describes alarm status, not the products' earlier conditions. It doesn't replace the responsible food-safety decision or establish an acceptable exposure history.

15 **Noah:** So I should leave the earlier product conditions unresolved, rather than assume that a cleared display would make the whole episode irrelevant.

16 **Imani:** Exactly. The [8] _____ isn't established by this extract. Keep the missing product data, duration, and cause visible alongside the known display reading.

17 **Noah:** My message says display eight Celsius at 06:10; no product readings supplied; alarm and defrost entries coincide; duration and cause unconfirmed.

18 **Imani:** That's a useful [9] _____. Send any relevant updates through the site process as they become available, keeping the product question separate from the equipment diagnosis.

19 **Noah:** The lead will handle the food decision under our procedures. I'll preserve the technical record and won't describe this call as approval to resume using the stock.

20 **Imani:** Right. [10] _____ belongs to that responsible process. We have clarified the evidence we have, not declared the food safe or resolved the equipment event.

CHECK THE GAPS

1. display value / **2.** product temperature / **3.** timestamp / **4.** event duration / **5.** food-safety lead / **6.** scheduled defrost / **7.** Alarm clearance / **8.** exposure history / **9.** escalation summary / **10.** Product disposition

Why these answers fit, page 139

F Check accuracy and scope

An alarm, a display value, and missing product data

Return to the case

Cafe manager Noah discusses a refrigeration record with technician Imani. A cabinet display recorded 8 degrees Celsius at 06:10. Product temperatures are not supplied. An alarm-history entry and a scheduled-defrost entry have the same time, but event cause and duration are unconfirmed. Noah has already alerted the responsible food-safety lead under the site procedures; this discussion occurs alongside that escalation and does not delay protective action. He asks how the records relate to the food-safety question. The display value and coincident entries do not authorize food release or establish a harmless defrost event.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Who should receive the food-release question?

A. Whoever recognizes the cabinet brand
B. The person who wants service to resume fastest
C. The alarm display as sole authority
D. The responsible food-safety lead using site procedures

2. Which statement improperly fills a data gap?

A. Product temperatures were not supplied
B. The duration remains unconfirmed
C. The food was warm for only one minute
D. The display value was recorded at 06:10

Answers and explanations, page 141

Listen for the distinction

Switch roles for turns 11-20. Contrast a scheduled-defrost entry with established cause, and alarm clearance with product disposition.

Give the record, not an improvised release decision

Operational pressure can turn a request for information into a request for reassurance. Identify exactly what the record contains and what is missing. Refer the product decision to the responsible food-safety lead while preserving the technical event as unresolved.

F Say it / transfer to a new case

Escalate the record without a safety claim

Complete the handoff to the responsible lead. Preserve the display record and the missing product information.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

8 / supplied / lead / 06:10

01 **Manager:** The display recorded [1] _____ degrees Celsius.

02 **Technician:** The timestamp was [2] _____.

03 **Manager:** Product temperatures were not [3] _____.

04 **Technician:** The responsible food-safety [4] _____ must decide under site procedures.

CHECK THE GAPS

1. 8 / **2.** 06:10 / **3.** supplied / **4.** lead

Why these answers fit, page 141

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "Suppose a later update says the alarm has disappeared. Would that alone answer whether the affected stock can be used?" Imani: [missing response] Noah: "So I should leave the earlier product conditions unresolved, rather than assume that a cleared display would make the whole episode irrelevant." Which response belongs in the gap?

A. Then product temperature is missing. Keep that gap explicit; don't copy the display number into a field that would imply the actual food was measured.

B. Then event duration is unconfirmed. A timestamp doesn't mean a one-minute event, and we shouldn't invent an interval to make the report look complete.

C. No. Alarm clearance describes alarm status, not the products' earlier conditions. It doesn't replace the responsible food-safety decision or establish an acceptable exposure history.

D. Yes. The display value is eight degrees Celsius at 06:10. It describes the equipment record, not a verified temperature for every stored item.

Answers and explanations, page 142

Review 3 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Reporting moisture, symptoms, and service findings" and "Discussing refrigeration alarms and temperature records". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Pipe or tube associated with carrying condensate away. Which term fits?

A. condensation

B. condensate line

C. dry observation

D. chronology

2. A colleague needs the term for this meaning: Defrost event listed in the control schedule or record. Which term fits?

A. scheduled defrost

B. refrigerated cabinet

C. alarm acknowledgment

D. exposure history

3. Rebuild the exchange from "Five kilowatts of what?". Jules: "I haven't received the indoor and outdoor pairings. Is a brochure for the outdoor unit enough if the installer chooses the indoor one later?" Farah: [missing response] Jules: "One summary includes a refrigerant name. Could I use that as evidence that the two proposals are interchangeable replacements?" Which response belongs in the gap?

A. Right. We need a like-for-like comparison using corresponding ratings and conditions. None of the missing values should be filled with the figure from the other proposal.

B. No. The equipment combination matters to the stated performance. We need to know which components the proposal actually pairs, not assume any indoor unit gives the same result.

C. Let's check five kilowatts of what. Proposal A labels it cooling capacity. B uses a different quantity label even though its number and unit look the same.

D. A refrigerant designation identifies a fluid or blend, not an equivalent system. It must be checked against the actual equipment specification and doesn't settle all compatibility requirements.

Review 3 / Meaning and context / continued

4. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "My message says display eight Celsius at 06:10; no product readings supplied; alarm and defrost entries coincide; duration and cause unconfirmed." Imani: [missing response] Noah: "The lead will handle the food decision under our procedures. I'll preserve the technical record and won't describe this call as approval to resume using the stock." Which response belongs in the gap?

A. It contains a scheduled defrost entry at the same recorded time as the alarm. That association needs assessment; it isn't yet a confirmed causal explanation.

B. That's a useful escalation summary. Send any relevant updates through the site process as they become available, keeping the product question separate from the equipment diagnosis.

C. Then product temperature is missing. Keep that gap explicit; don't copy the display number into a field that would imply the actual food was measured.

D. The shared timestamp only tells us the entries coincide. It doesn't establish the cause, prove the event was harmless, or determine what happened to the products.

Answers and explanations, page 158

A Read the situation

LESSON 07 / COORDINATING DUCTWORK, ACCESS, AND CHANGES

Acknowledge an appearance preference, explain unassessed consequences, and present two bounded choices without promising price or timing.

> **Moving a grille is a coordinated change**
>
> Office client Priya asks project coordinator Daniel to move a supply grille because it visually conflicts with a pendant light. The proposal has not assessed ductwork, ceiling restoration, service access, price, or timing. Priya may retain the current reviewed layout or request a coordinated change proposal. Daniel must explain the choice without treating a small visible move as a minor technical task or guaranteeing unchanged cost and completion.

People: Priya, Office client; Daniel, Project coordinator.

Read the record

Choose one answer per question using the facts above.

1. Why does Priya want the grille moved?

A. A measured airflow failure is established
B. A new layout has already been accepted
C. A maintenance inspection ordered the move
D. It visually conflicts with a pendant light

2. Which consequences have been assessed?

A. None of the listed change effects
B. Ductwork and price only
C. Service access and timing only
D. All effects with a fixed quote

3. What are the stated choices?

A. Accept a guessed price or waive review
B. Change the refrigerant or replace the system
C. Retain the reviewed layout or request a coordinated change proposal
D. Move it immediately or cancel the whole project

Answers and explanations, page 143

B Vocabulary in use

CORE TERMS

supply grille relocation
Proposed change to the position of a supply-air grille.

assess a supply grille relocation

visual conflict
Appearance issue between elements in the same view.

describe the visual conflict

ceiling void
Space above the visible ceiling and below the structure above.

coordinate the ceiling void

duct connection
Interface joining a duct to another component.

assess the duct connection

rigid duct
Air-conveying duct with a relatively fixed form.

distinguish rigid duct

ceiling restoration
Reinstatement of ceiling finishes after defined work.

assess ceiling restoration

pendant light
Light fitting suspended from the ceiling.

coordinate the pendant light

reflected ceiling plan
Drawing of ceiling elements represented as viewed in reflection from below.

review the reflected ceiling plan

branch duct
Duct section branching from a larger distribution route.

review the branch duct

flexible duct
Air-conveying duct with a flexible construction.

identify the flexible duct

service clearance
Space required for relevant maintenance or access.

preserve service clearance

access provision
Means included in the design for reaching services.

review the access provision

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

supply grille relocation SUPPLY GRILLE RELOCATION

assess a supply grille relocation

Keep the meaning precise: Proposed change to the position of a supply-air grille.

branch duct BRANCH DUCT

review the branch duct

Keep the meaning precise: Duct section branching from a larger distribution route.

B Precision and professional shorthand

EXTEND YOUR RANGE

reviewed layout
Arrangement that has undergone the stated project review.

retain the reviewed layout

change proposal
Defined suggestion for altering the existing arrangement.

request a change proposal

coordination review
Evaluation of the change across affected elements and trades.

arrange a coordination review

scope impact
Effect of a change on the work required.

identify the scope impact

cost impact
Effect of a change on price or expenditure.

assess the cost impact

schedule impact
Effect of a change on timing.

assess the schedule impact

design preference
Client choice concerning the desired arrangement or appearance.

record the design preference

technical feasibility
Whether the proposed arrangement can satisfy relevant technical conditions.

establish technical feasibility

revised layout
Updated arrangement proposed or accepted after a change.

review the revised layout

unassessed consequence
Effect not yet evaluated.

keep unassessed consequences visible

instruction to proceed
Direction authorizing agreed work to begin.

distinguish an instruction to proceed

decision record
Record of the option selected and its conditions.

confirm the decision record

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

access provision ACCESS PROVISION

review the access provision

Keep the meaning precise: Means included in the design for reaching services.

cost impact COST IMPACT

assess the cost impact

Keep the meaning precise: Effect of a change on price or expenditure.

C Phrases that move the work forward

Acknowledge the preference

I understand why you want the grille clear of the pendant light.

Name the request

You are asking to relocate the supply grille.

Avoid minimizing hidden work

The visible move may involve work above the ceiling.

Identify the unassessed items

Ductwork, restoration, access, price, and timing still need assessment.

Separate appearance and fault

The concern is visual, not a confirmed performance defect.

State the first option

You can retain the current reviewed layout.

State the second option

You can request a coordinated change proposal.

Keep the choice bounded

Those are the two decisions available at this stage.

Avoid a price promise

I cannot confirm unchanged cost before review.

Avoid a timing promise

The schedule effect has not been established.

Keep access in scope

The proposal must account for service access.

Clarify the request status

Asking for a proposal does not authorize the relocation.

Preserve technical review

The new position needs the relevant coordination.

Name restoration explicitly

Ceiling restoration must be included in the assessment.

Read back the decision

You are requesting assessment, not approving installation.

Close with the next deliverable

We will bring back a defined proposal with its consequences.

Precision in real exchanges

Purpose: Identify the unassessed items
Useful wording: Ductwork, restoration, access, price, and timing still need assessment.

Purpose: Separate appearance and fault
Useful wording: The concern is visual, not a confirmed performance defect.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which response explains the choice accurately?

A. The move is already approved because the client prefers it

B. Only the grille position matters; access can be ignored

C. Retain the reviewed layout or request a coordinated proposal with impacts assessed

D. The move is free because it is visually small

2. Which item must not disappear from review?

A. A promised unchanged date

B. Service access

C. A guessed refrigerant change

D. An invented system failure

3. A colleague needs the term for this meaning: Drawing of ceiling elements represented as viewed in reflection from below. Which term fits?

A. revised layout

B. rigid duct

C. reflected ceiling plan

D. coordination review

D Choose the response that fits / continued

4. Rebuild the exchange from "Moving a grille is a coordinated change". Priya: "What would need assessment beyond marking a new position on the ceiling plan? I hadn't thought about the connection above it." Daniel: [missing response] Priya: "Would restoring the old ceiling opening be part of that review too? I don't want a price that leaves the finishing work unexplained." Which response belongs in the gap?

A. Understood. We should call it a visual conflict, not an established performance defect. That keeps the reason for the request accurate when it reaches the design team.

B. Correct. The access provision needs review with the other requirements. Appearance is important, but it does not by itself establish whether a revised arrangement is technically acceptable.

C. It is a design preference we can record, but the visible size of the move does not tell us the amount of work involved above the ceiling.

D. The ductwork has not been assessed, nor have ceiling restoration, service access, price, or timing. A revised drawing must be supported by review of those related effects.

Answers and explanations, page 143

One useful distinction

The visible conflict concerns the grille and pendant light. No performance fault is established. Ductwork, restoration, access, price, and timing all need assessment before the proposed move can be treated as a defined change.

E Conversation

Moving a grille is a coordinated change

Moving a grille is a coordinated change

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

reviewed layout / change proposal / Ceiling restoration / access provision / decision record / ductwork / cost impact / schedule impact / design preference / visual conflict

01 **Priya:** Daniel, the supply grille looks awkward beside the pendant light. I'd like it moved. Is the small distance on the ceiling a small change to the work?

02 **Daniel:** It is a [1] _____ we can record, but the visible size of the move does not tell us the amount of work involved above the ceiling.

03 **Priya:** Yes, it's an appearance issue. I'm not reporting poor cooling or saying the grille has failed; I want the light and grille to sit better together.

04 **Daniel:** Understood. We should call it a [2] _____, not an established performance defect. That keeps the reason for the request accurate when it reaches the design team.

05 **Priya:** What would need assessment beyond marking a new position on the ceiling plan? I hadn't thought about the connection above it.

06 **Daniel:** The [3] _____ has not been assessed, nor have ceiling restoration, service access, price, or timing. A revised drawing must be supported by review of those related effects.

E Conversation / continued

07 **Priya:** Would restoring the old ceiling opening be part of that review too? I don't want a price that leaves the finishing work unexplained.

08 **Daniel:** Yes. [4] _____ must be assessed rather than treated as automatically included or unnecessary. We should not give you a price for only part of the actual change.

09 **Priya:** Can we also check service access? Moving something to a neater position wouldn't help if it makes the relevant services hard to reach.

10 **Daniel:** Correct. The [5] _____ needs review with the other requirements. Appearance is important, but it does not by itself establish whether a revised arrangement is technically acceptable.

Conversation continues on the next page.

Moving a grille is a coordinated change

CONVERSATION / CONTINUED

11 **Priya:** What decision can I make now? I don't have the information to design the duct route or choose a technically acceptable new position myself.

12 **Daniel:** You can retain the current [6] _____, or request a coordinated change proposal. Those are the two options at this stage; neither requires you to specify a duct route.

13 **Priya:** I'd like a proposal. Please don't treat that request as acceptance of an unknown extra cost or as permission to start the move.

14 **Daniel:** It does not. A [7] _____ will define the alternative and its consequences for review. Requesting it is separate from accepting the resulting work and commercial terms.

15 **Priya:** Can you keep the original finish date? The office move-in depends on it, though I realize the schedule effect hasn't been assessed yet.

16 **Daniel:** I cannot confirm that yet. The [8] _____ is one of the open questions, along with cost. Your timing concern should be recorded without being mistaken for a promise that no effect exists.

17 **Priya:** Then please record timing as a concern, not as your guarantee. And avoid calling the move free or minor before the full scope is known.

18 **Daniel:** Agreed. The [9] _____ will need a defined scope. We will include ductwork, access, restoration, and timing in the coordinated review rather than price only the visible grille.

Moving a grille is a coordinated change / continued

19 **Priya:** That's my decision: assess a coordinated proposal and return the consequences for review. I'm not instructing the team to relocate the grille now.

20 **Daniel:** That is the [10] _____: coordinated proposal requested; relocation not authorized. The current reviewed arrangement is not replaced by an assumed new position during this conversation.

CHECK THE GAPS

1. design preference / **2.** visual conflict / **3.** ductwork / **4.** Ceiling restoration / **5.** access provision / **6.** reviewed layout / **7.** change proposal / **8.** schedule impact / **9.** cost impact / **10.** decision record

Why these answers fit, page 144

F Check accuracy and scope

Moving a grille is a coordinated change

Return to the case

Office client Priya asks project coordinator Daniel to move a supply grille because it visually conflicts with a pendant light. The proposal has not assessed ductwork, ceiling restoration, service access, price, or timing. Priya may retain the current reviewed layout or request a coordinated change proposal. Daniel must explain the choice without treating a small visible move as a minor technical task or guaranteeing unchanged cost and completion.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What does requesting a proposal authorize here?

A. Assessment of the change, not its installation
B. Immediate relocation
C. Automatic acceptance of any price
D. Waiving ceiling restoration

2. Which phrase overstates certainty?

A. We need to assess the ductwork
B. The conflict is visual
C. You can retain the current layout
D. We can move it with no cost or timing effect

Answers and explanations, page 146

Listen for the distinction

Switch roles for turns 11-20. Stress retain the layout or request a proposal, with no approval to relocate.

Make the choice clear without minimizing the work

A visible component can be connected to less visible work above the ceiling. Acknowledge the design preference, list the specific unassessed matters, and offer the stated choices. A request to assess a move is not approval to install it or an agreement to absorb its cost.

F Say it / transfer to a new case

Offer two clear paths

Complete the choice discussion. Keep the existing layout and the request for assessment separate from installation approval.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

relocation / assessment / layout / proposal

01 **Coordinator:** You can retain the reviewed [1] _____.

02 **Client:** Or request a coordinated change [2] _____.

03 **Coordinator:** Cost and timing still need [3] _____.

04 **Client:** This request does not authorize [4] _____.

CHECK THE GAPS

1. layout / **2.** proposal / **3.** assessment / **4.** relocation

Why these answers fit, page 146

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Moving a grille is a coordinated change". Priya: "I'd like a proposal. Please don't treat that request as acceptance of an unknown extra cost or as permission to start the move." Daniel: [missing response] Priya: "Can you keep the original finish date? The office move-in depends on it, though I realize the schedule effect hasn't been assessed yet." Which response belongs in the gap?

A. Understood. We should call it a visual conflict, not an established performance defect. That keeps the reason for the request accurate when it reaches the design team.

B. It is a design preference we can record, but the visible size of the move does not tell us the amount of work involved above the ceiling.

C. It does not. A change proposal will define the alternative and its consequences for review. Requesting it is separate from accepting the resulting work and commercial terms.

D. The ductwork has not been assessed, nor have ceiling restoration, service access, price, or timing. A revised drawing must be supported by review of those related effects.

Answers and explanations, page 147

A Read the situation

LESSON 08 / HANDING OVER COMMISSIONING AND MAINTENANCE INFORMATION

Separate document receipt from completed verification and distinguish assigned correction work from an unassigned scheduling responsibility.

> **Report received, seasonal check still open**
>
> Client Ben receives a commissioning report for a fictional office system. It lists a deferred seasonal check, and the maintenance sheet names the wrong model. Coordinator Nia accepts responsibility for document correction and will update Ben on Friday. The seasonal-check date is not booked, and responsibility for scheduling it has not been confirmed. The handover must preserve that ownership gap. No overall performance conclusion, completed seasonal result, or change to warranty terms is supplied.

People: Ben, Client; Nia, Coordinator.

Read the record

Choose one answer per question using the facts above.

1. What does the commissioning report list?

A. A completed seasonal result
B. A confirmed Friday test booking
C. No outstanding verification
D. A deferred seasonal check

2. What responsibility does Nia accept?

A. An automatic warranty extension
B. Document correction and a Friday update
C. Confirmed ownership of scheduling every check
D. A guaranteed system-performance result

3. What remains unresolved about the seasonal check?

A. Whether any report was received
B. Whether Nia agreed to correct the sheet
C. Both the date and the scheduling owner
D. Only the spelling of Friday

Answers and explanations, page 148

B Vocabulary in use

CORE TERMS

commissioning

Process of verifying and documenting system performance against relevant project requirements.

distinguish commissioning from document delivery

commissioning report

Record of commissioning activities, results, and outstanding matters.

receive the commissioning report

deferred check

Verification postponed until a later appropriate time or condition.

record the deferred check

seasonal check

Verification associated with relevant seasonal operating conditions.

schedule the seasonal check

maintenance sheet

Document describing relevant maintenance information for equipment.

correct the maintenance sheet

model-specific guidance

Information applicable to the actual identified product model.

obtain model-specific guidance

document mismatch

Difference between supplied information and the relevant equipment.

identify the document mismatch

correction owner

Person responsible for resolving a record error.

name the correction owner

scheduling owner

Person responsible for arranging the required attendance or check.

confirm the scheduling owner

ownership gap

Task for which responsibility has not been confirmed.

preserve the ownership gap

B Vocabulary in use / continued

unbooked check

Required verification with no confirmed appointment.

retain the unbooked check

update commitment

Promise to provide further information at a stated time.

honor the update commitment

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

commissioning COMMISSIONING

distinguish commissioning from document delivery

Keep the meaning precise: Process of verifying and documenting system performance against relevant project requirements.

model-specific guidance MODEL-SPECIFIC GUIDANCE

obtain model-specific guidance

Keep the meaning precise: Information applicable to the actual identified product model.

B Precision and professional shorthand

EXTEND YOUR RANGE

outstanding verification

Check not yet completed or accepted.

track outstanding verification

handover register

Record of delivered information and remaining handover matters.

update the handover register

operating record

Documented information about actual operation.

retain the operating record

performance result

Established outcome from relevant measurement or verification.

distinguish a performance result

acceptance criterion

Requirement used to judge whether an outcome meets the agreed standard.

identify the acceptance criterion

test condition

Defined state under which a verification is carried out.

record the test condition

deferral reason

Explanation for postponing an activity.

preserve the deferral reason

closeout status

Current position of remaining completion matters.

state the closeout status

maintenance requirement

Relevant action or condition specified for upkeep.

verify the maintenance requirement

warranty terms

Actual conditions and coverage stated in the relevant warranty.

preserve warranty terms

document receipt

Confirmation that information has been delivered.

acknowledge document receipt

responsibility readback

Repetition of assigned and unassigned tasks for clarity.

give a responsibility readback

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

update commitment UPDATE COMMITMENT

honor the update commitment

Keep the meaning precise: Promise to provide further information at a stated time.

acceptance criterion ACCEPTANCE CRITERION

identify the acceptance criterion

Keep the meaning precise: Requirement used to judge whether an outcome meets the agreed standard.

C Phrases that move the work forward

Acknowledge the report
You have received the commissioning report.

Identify the document error
The maintenance sheet names the wrong model.

Commit to an update
I will update you on Friday.

State the booking gap
The seasonal-check date is not booked.

Avoid implied ownership
My document task does not settle who schedules the check.

Separate receipt and verification
Receiving the report does not complete the deferred check.

Keep warranty separate
This discussion does not change the actual warranty terms.

Read back responsibilities
Nia owns correction; seasonal scheduling ownership remains open.

Name the exception
The report lists a deferred seasonal check.

Accept a specific task
I will take responsibility for document correction.

Keep the date limited
Friday is an update date, not a booked seasonal check.

State the ownership gap
Responsibility for arranging that check is not confirmed.

Preserve model specificity
The replacement sheet must match the actual model.

Avoid an overall assurance
We have no overall performance conclusion to add here.

Name the next coordination issue
We still need a confirmed scheduling owner and date.

Close with accurate status
Documents are received, corrections and deferred verification remain visible.

Precision in real exchanges

Purpose: Accept a specific task
Useful wording: I will take responsibility for document correction.

Purpose: Commit to an update
Useful wording: I will update you on Friday.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which summary preserves all responsibilities?

A. The client automatically owns the seasonal check

B. Nia corrects documents and updates Friday; seasonal-check owner and date remain open

C. Nia has booked every check for Friday

D. Report receipt proves full performance

2. What does Friday identify?

A. Completion of every open item

B. Nia's promised update

C. A confirmed seasonal test date

D. A warranty start date supplied by this case

3. A colleague needs the term for this meaning: Verification associated with relevant seasonal operating conditions. Which term fits?

A. operating record

B. scheduling owner

C. seasonal check

D. maintenance requirement

D Choose the response that fits / continued

4. Rebuild the exchange from "Report received, seasonal check still open". Ben: "Who is arranging the seasonal check? I also can't find a booked date, so both responsibility and scheduling seem unresolved." Nia: [missing response] Ben: "I nearly assigned that to you because you're handling the wrong sheet. That would add a responsibility you haven't actually accepted." Which response belongs in the gap?

A. The scheduling owner has not been confirmed, and no date is booked. I should not imply that my document-correction task resolves either of those questions.

B. That responsibility readback is accurate. It makes the commitment and both remaining scheduling questions visible instead of assigning the whole handover to one person by implication.

C. Correct. The warranty terms are not changed by this conversation. Neither an update date nor an outstanding check creates a new warranty promise here.

D. No. The deferred check is still outstanding. The report records its status; delivery of that report does not carry out the verification or supply its result.

Answers and explanations, page 148

> **One useful distinction**
>
> The report has been received, but the seasonal check is deferred. The wrong-model maintenance sheet needs correction. Neither delivery nor a corrected document would itself establish a completed seasonal result or overall system performance.

E Conversation

Report received, seasonal check still open

Report received, seasonal check still open

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

handover register / performance result / update commitment / responsibility readback / document mismatch / scheduling owner / deferred check / warranty terms / model-specific guidance / ownership gap

01 **Ben:** Nia, the commissioning report arrived, but it lists a deferred seasonal check. The maintenance sheet also names a different model from our actual system.

02 **Nia:** Those are two distinct handover matters. I accept the [1] _____ on the maintenance sheet and will take responsibility for correcting that information.

03 **Ben:** Thanks for taking the document correction. I want to separate that from the outstanding check rather than mark everything complete on receipt of the report.

04 **Nia:** No. The [2] _____ is still outstanding. The report records its status; delivery of that report does not carry out the verification or supply its result.

05 **Ben:** Who is arranging the seasonal check? I also can't find a booked date, so both responsibility and scheduling seem unresolved.

06 **Nia:** The [3] _____ has not been confirmed, and no date is booked. I should not imply that my document-correction task resolves either of those questions.

07 **Ben:** I nearly assigned that to you because you're handling the wrong sheet. That would add a responsibility you haven't actually accepted.

08 **Nia:** Exactly. We need to preserve the [4] _____ for the seasonal check. It needs an explicit assignment, not an assumption based on who is speaking at handover.

09 **Ben:** What date can I put down for the next update? I need a communication commitment even while the technical appointment remains unbooked.

10 **Nia:** I will correct the document information and make an [5] _____ for Friday. That gives you a communication date without presenting an unbooked seasonal check as scheduled.

Conversation continues on the next page.

Report received, seasonal check still open

CONVERSATION / CONTINUED

11 **Ben:** Please match the corrected sheet to the actual model. A similar product leaflet would still leave the maintenance information uncertain.

12 **Nia:** Agreed. We need [6] _____ for the actual equipment. Correcting the sheet is a defined records task; it is not evidence that the seasonal verification has occurred.

13 **Ben:** Would report received support a statement that the system has passed every operating condition? The deferred check suggests that would be too broad.

14 **Nia:** No. We have no overall [7] _____ to add from these facts. The deferred item remains part of the record, and we should not invent a successful result for it.

15 **Ben:** I also want the summary to preserve the actual warranty terms, without implying the open check creates a new warranty promise.

16 **Nia:** Correct. The [8] _____ are not changed by this conversation. Neither an update date nor an outstanding check creates a new warranty promise here.

17 **Ben:** Let me check the responsibilities: you own document correction and the Friday update. The seasonal-check owner and date are both still unconfirmed.

18 **Nia:** That [9] _____ is accurate. It makes the commitment and both remaining scheduling questions visible instead of assigning the whole handover to one person by implication.

19 **Ben:** I'll acknowledge receipt and keep the unresolved items separate. Friday goes in the update column, not the seasonal-test appointment column.

20 **Nia:** Thank you. The [10] _____ should preserve the report receipt, wrong-model correction, deferred check, and open scheduling responsibility and date. That is a clear close to today's conversation, with the unfinished work still visible.

CHECK THE GAPS

1. document mismatch / **2.** deferred check / **3.** scheduling owner / **4.** ownership gap / **5.** update commitment / **6.** model-specific guidance / **7.** performance result / **8.** warranty terms / **9.** responsibility readback / **10.** handover register

Why these answers fit, page 149

F Check accuracy and scope

Report received, seasonal check still open

Return to the case

Client Ben receives a commissioning report for a fictional office system. It lists a deferred seasonal check, and the maintenance sheet names the wrong model. Coordinator Nia accepts responsibility for document correction and will update Ben on Friday. The seasonal-check date is not booked, and responsibility for scheduling it has not been confirmed. The handover must preserve that ownership gap. No overall performance conclusion, completed seasonal result, or change to warranty terms is supplied.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which inference is unsupported?

A. Scheduling ownership is unconfirmed
B. Correcting the maintenance sheet completes the seasonal check
C. The sheet names the wrong model
D. The report was received

2. What should the handover keep open?

A. Only a date, because Nia's document task assigns her the seasonal check
B. A Friday test appointment inferred from the update commitment
C. Correct model guidance, deferred verification, scheduling owner, and date
D. Only the leaflet, because receiving the report completes verification

Answers and explanations, page 151

Listen for the distinction

Switch roles for turns 11-20. Keep Friday as an update and report receipt separate from a completed performance check.

Do not let one named owner hide a different unassigned task

A coordinator can own a document correction without owning every remaining technical action. A useful handover names the received records, the unresolved checks, and who is responsible for each. State an ownership gap directly rather than letting a polite promise to update imply it is settled.

F Say it / transfer to a new case

Assign only the responsibility actually accepted

Complete the handover readback. Keep the document owner, update date, and unassigned seasonal-check task distinct.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

unbooked / Friday / Nia / confirmed

01 **Client:** [1] _____ owns document correction.

02 **Coordinator:** The update is promised for [2] _____.

03 **Client:** The seasonal check is still [3] _____.

04 **Coordinator:** Its scheduling owner is not yet [4] _____.

CHECK THE GAPS

1. Nia / **2.** Friday / **3.** unbooked / **4.** confirmed

Why these answers fit, page 151

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Report received, seasonal check still open". Ben: "Would report received support a statement that the system has passed every operating condition? The deferred check suggests that would be too broad." Nia: [missing response] Ben: "I also want the summary to preserve the actual warranty terms, without implying the open check creates a new warranty promise." Which response belongs in the gap?

A. Agreed. We need model-specific guidance for the actual equipment. Correcting the sheet is a defined records task; it is not evidence that the seasonal verification has occurred.

B. I will correct the document information and make an update commitment for Friday. That gives you a communication date without presenting an unbooked seasonal check as scheduled.

C. No. We have no overall performance result to add from these facts. The deferred item remains part of the record, and we should not invent a successful result for it.

D. Exactly. We need to preserve the ownership gap for the seasonal check. It needs an explicit assignment, not an assumption based on who is speaking at handover.

Answers and explanations, page 152

Review 4 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Coordinating ductwork, access, and changes" and "Handing over commissioning and maintenance information". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Appearance issue between elements in the same view. Which term fits?

A. flexible duct

B. change proposal

C. technical feasibility

D. visual conflict

2. A colleague needs the term for this meaning: Person responsible for resolving a record error. Which term fits?

A. outstanding verification

B. commissioning

C. correction owner

D. deferral reason

3. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "I'll keep the lead updated with those gaps. Our opening time is close, but the technical call isn't a substitute for the product decision." Imani: [missing response] Noah: "Please distinguish the scheduled item from the actual explanation. I want the handoff to say what the log contains without implying we've diagnosed the event." Which response belongs in the gap?

A. The shared timestamp only tells us the entries coincide. It doesn't establish the cause, prove the event was harmless, or determine what happened to the products.

B. That's a useful escalation summary. Send any relevant updates through the site process as they become available, keeping the product question separate from the equipment diagnosis.

C. Correct. The food-safety lead must decide under your site procedures. Follow that process while the equipment information is clarified; don't wait for a reassuring guess from the display.

D. Then product temperature is missing. Keep that gap explicit; don't copy the display number into a field that would imply the actual food was measured.

Review 4 / Meaning and context / continued

4. Rebuild the exchange from "Report received, seasonal check still open". Ben: "Let me check the responsibilities: you own document correction and the Friday update. The seasonal-check owner and date are both still unconfirmed." Nia: [missing response] Ben: "I'll acknowledge receipt and keep the unresolved items separate. Friday goes in the update column, not the seasonal-test appointment column." Which response belongs in the gap?

A. No. The deferred check is still outstanding. The report records its status; delivery of that report does not carry out the verification or supply its result.

B. That responsibility readback is accurate. It makes the commitment and both remaining scheduling questions visible instead of assigning the whole handover to one person by implication.

C. The scheduling owner has not been confirmed, and no date is booked. I should not imply that my document-correction task resolves either of those questions.

D. Exactly. We need to preserve the ownership gap for the seasonal check. It needs an explicit assignment, not an assumption based on who is speaking at handover.

Answers and explanations, page 161

Language for the next case

Total cooling hides the split

sensible load

The supplied ten-kilowatt sensible load concerns temperature change; it must be compared with sensible capacity, not total capacity alone.

Theo: We need the split. The room's sensible load is ten kilowatts, while A's sensible capacity is nine. Matching the totals hides a one-kilowatt shortfall.

latent load

The two-kilowatt latent load concerns the moisture-removal part of this cooling duty; it is not a temperature value in degrees.

Theo: Correct. That's the latent load. Sensible cooling concerns temperature change; latent cooling here concerns condensation of water vapor removed from the air.

total capacity

Twelve kilowatts total equals nine sensible plus three latent here; extra latent capacity does not erase the stated sensible-capacity shortfall.

Theo: Yes, but the total capacity doesn't make the two parts interchangeable. The extra latent capacity doesn't establish ten kilowatts of sensible cooling at this condition.

sensible heat ratio

SHR is sensible capacity divided by total capacity, nine divided by twelve or 0.75 for A; it is not output divided by electrical input.

Theo: No. The sensible heat ratio describes the sensible share of cooling, not electrical efficiency. We haven't been given input power for an efficiency calculation.

comparison basis

The case gives consistent net capacities at the same conditions; B's 10.4 sensible and 2.6 latent exceed the supplied ten and two loads respectively.

Theo: Yes, on the stated comparison basis. B's SHR is ten point four divided by thirteen, or zero point eight.

load profile

Different combinations of temperature and moisture loads require different comparisons; the highest sensible fraction is not automatically best for every space.

Theo: Exactly. A room with a different load profile could need a different balance. We compare the relevant components and conditions, not a ratio as a universal quality score.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

Total cooling hides the split

Fictional comparison on a consistent net-capacity basis at the same stated operating conditions: room load is 10 kW sensible plus 2 kW latent. Option A supplies 12 kW total, including 9 kW sensible; B supplies 13 kW total, including 10.4 kW sensible. Total = sensible + latent; sensible heat ratio (SHR) = sensible / total. Actual equipment selection and installation are not approved.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

sensible load / sensible heat ratio / total capacity / comparison basis / load profile / latent load

01 **Mina:** Option A says twelve kilowatts, exactly the room's total load. Can I mark it as meeting the requirement before we compare the price?

02 **Theo:** We need the split. The room's [1] _____ is ten kilowatts, while A's sensible capacity is nine. Matching the totals hides a one-kilowatt shortfall.

03 **Mina:** And the other two kilowatts in the room load concern moisture removal. They aren't another two degrees of temperature reduction.

04 **Theo:** Correct. That's the [2] _____. Sensible cooling concerns temperature change; latent cooling here concerns condensation of water vapor removed from the air.

05 **Mina:** If A has twelve total and nine sensible, its remaining capacity is three latent. That's one more than the latent load.

06 **Theo:** Yes, but the [3] _____ doesn't make the two parts interchangeable. The extra latent capacity doesn't establish ten kilowatts of sensible cooling at this condition.

Conversation / continued

07 **Mina:** The worksheet also lists SHR. For A, nine divided by twelve gives zero point seven five. Does that mean seventy-five percent efficiency?

08 **Theo:** No. The [4] _____ describes the sensible share of cooling, not electrical efficiency. We haven't been given input power for an efficiency calculation.

09 **Mina:** B gives thirteen total and ten point four sensible. Subtracting leaves two point six latent, so both supplied load components are covered numerically.

10 **Theo:** Yes, on the stated [5] _____. B's SHR is ten point four divided by thirteen, or zero point eight.

Conversation continues on the next page.

Total cooling hides the split

CONVERSATION / CONTINUED

11 **Mina:** A bigger sensible ratio doesn't automatically mean a better unit, then. The right split depends on what the space needs, not the highest SHR.

12 **Theo:** Exactly. A room with a different [6] _____ could need a different balance. We compare the relevant components and conditions, not a ratio as a universal quality score.

13 **Mina:** Can the brochure's headline capacity replace these values? It may have been rated at a different outdoor temperature and entering-air condition.

14 **Theo:** Not without checking the conditions. Use the performance data for the comparison being made, including airflow and the equipment combination, rather than swapping in a favorable headline.

15 **Mina:** The case says net capacity. If another sheet says gross coil capacity, should I use it directly because both are still in kilowatts?

16 **Theo:** No. Gross and net values can have different treatment of fan heat. Reconcile the stated basis before comparing them; identical units don't guarantee equivalent boundaries.

17 **Mina:** Then B clears this worksheet's sensible and latent checks, but we still haven't selected equipment for the actual building.

18 **Theo:** Right. Part-load behavior, controls, conditions, and the rest of the design assessment remain relevant. This arithmetic is not approval to purchase or install a system.

19 **Mina:** I'll record A as nine sensible and three latent, with a one-kilowatt sensible shortfall. B is ten point four sensible and two point six latent.

20 **Theo:** Include the load split, rating conditions, and net-capacity basis. That explains why a twelve-kilowatt total alone was not enough to settle the comparison.

CHECK THE GAPS

1. sensible load / **2.** latent load / **3.** total capacity / **4.** sensible heat ratio / **5.** comparison basis / **6.** load profile

Why these answers fit, page 164

Check the components, not just the total

Fictional like-for-like net-capacity comparison: load 6 kW sensible and 2 kW latent. Option C: 8 kW total, 6.4 kW sensible. Option D: 9 kW total, 6.3 kW sensible. Use latent = total - sensible and SHR = sensible / total. Compare only these stated figures.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

1.6 / 0.4 / D / 0.7

01 **Manager:** C's latent capacity is [1] _____ kW.

02 **Estimator:** C therefore has a latent shortfall of [2] _____ kW.

03 **Manager:** Option [3] _____ covers both supplied component loads numerically.

04 **Estimator:** D's sensible heat ratio is [4] _____ .

CHECK THE GAPS

1. 1.6 / **2.** 0.4 / **3.** D / **4.** 0.7

Why these answers fit, page 164

Keep the distinction audible

Compare sensible and latent cooling separately instead of relying on one total rating.

Language for the next case

Circulation is not all outdoor air

total supply

Six hundred cubic meters per hour divided by a three-hundred-cubic-meter room gives two total-supply air changes per hour, including recirculated air.

Leah: Not as written. Dividing the total supply by room volume gives two air changes per hour, but the supply includes recirculated air.

air changes per hour

ACH divides a consistently expressed hourly airflow by room volume; the total-supply result is two and the outdoor-air result is 0.5.

Leah: Yes: two total-supply and zero point five outdoor-air air changes per hour. The shared unit doesn't mean the two figures represent the same air source.

time basis

A cubic meter contains one thousand liters and an hour has 3,600 seconds; 600 cubic meters per hour is about 166.7 liters per second.

Leah: Yes. Keep the time basis straight: multiply by one thousand, then divide by thirty-six hundred. Outdoor flow is about forty-one point seven liters per second.

outdoor-air flow

Only 150 cubic meters per hour is outdoor air in this case; the other 450 is recirculated, not an additional outdoor-air contribution.

Leah: Correct. The outdoor-air flow is one hundred fifty cubic meters per hour. The other four hundred fifty is recirculated within the stated arrangement.

outdoor-air fraction

Outdoor flow divided by total supply is 150/600 = 0.25, or 25 percent; this denominator is flow, not room volume.

Leah: Exactly. The outdoor-air fraction is one quarter, or twenty-five percent. Don't confuse that fraction with zero point five air changes per hour.

air mixing

An air-change quantity describes delivered volume relative to room volume; it does not guarantee complete replacement of every portion of room air in a fixed interval.

Leah: No. Actual air mixing and distribution matter, and much of this supply is recirculated. One room-volume of delivery doesn't sweep out every original molecule.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

Circulation is not all outdoor air

Fictional steady-flow room report: volume 300 cubic meters; total supply 600 cubic meters per hour, comprising 150 outdoor and 450 recirculated. Assume these flows serve this room on the same volume basis. Ignore other air paths for this arithmetic only. Air changes per hour = hourly flow / room volume. No ventilation requirement, filtration effectiveness, or health assurance is supplied.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

outdoor-air flow / time basis / outdoor-air fraction / air changes per hour / air mixing / total supply

01 **Omar:** The report gives six hundred cubic meters an hour for a three-hundred-cubic-meter room. I wrote two complete fresh-air replacements every hour. Is that accurate?

02 **Leah:** Not as written. Dividing the [1] _____ by room volume gives two air changes per hour, but the supply includes recirculated air.

03 **Omar:** Only one hundred fifty of the six hundred comes from outside. I shouldn't call all six hundred fresh outdoor air just because it enters the room.

04 **Leah:** Correct. The [2] _____ is one hundred fifty cubic meters per hour. The other four hundred fifty is recirculated within the stated arrangement.

05 **Omar:** Then the outdoor-air calculation is one hundred fifty divided by three hundred, or zero point five. I need to label both results.

06 **Leah:** Yes: two total-supply and zero point five outdoor-air [3] _____. The shared unit doesn't mean the two figures represent the same air source.

Conversation / continued

07 **Omar:** If I want the percentage of supply that is outdoor air, I divide one hundred fifty by six hundred rather than by room volume.

08 **Leah:** Exactly. The [4] _____ is one quarter, or twenty-five percent. Don't confuse that fraction with zero point five air changes per hour.

09 **Omar:** The equipment summary uses liters per second instead. Six hundred cubic meters an hour would be about one hundred sixty-six point seven liters per second?

10 **Leah:** Yes. Keep the [5] _____ straight: multiply by one thousand, then divide by thirty-six hundred. Outdoor flow is about forty-one point seven liters per second.

Conversation continues on the next page.

Circulation is not all outdoor air

CONVERSATION / CONTINUED

11 **Omar:** Does two total air changes mean every bit of the original room air has gone after half an hour? The word replacement makes it sound that way.

12 **Leah:** No. Actual [6] _____ and distribution matter, and much of this supply is recirculated. One room-volume of delivery doesn't sweep out every original molecule.

13 **Omar:** Can the recirculated portion still contribute to particle removal if the system has appropriate filtration? I don't want to say it has no possible benefit.

14 **Leah:** It can, but this report gives no filtration-effectiveness evidence for a clean-air calculation. Don't count all recirculation as clean air automatically, or call it outdoor air.

15 **Omar:** So we can't combine the two total-supply changes and the half outdoor change and advertise two and a half. The outdoor portion is already inside the total.

16 **Leah:** Exactly. That would count the outdoor flow twice. Six hundred already equals one hundred fifty plus four hundred fifty under the case's stated assumptions.

17 **Omar:** And none of these calculations says the room meets its actual ventilation requirement or makes the air safe for a particular use.

18 **Leah:** Correct. Occupancy, the applicable design requirements, distribution, filtration, and other conditions need assessment. This record alone isn't a compliance or health determination.

19 **Omar:** I'll replace fresh-air replacements with two total-supply ACH, including zero point five outdoor-air ACH, and state the twenty-five-percent outdoor fraction.

20 **Leah:** Keep the room volume and original flow units alongside those figures. That lets the reader check both the arithmetic and exactly which air quantity each result describes.

CHECK THE GAPS

1. total supply / **2.** outdoor-air flow / **3.** air changes per hour / **4.** outdoor-air fraction / **5.** time basis / **6.** air mixing

Why these answers fit, page 166

Keep the two air-change rates apart

Fictional room volume 240 cubic meters; total supply 720 cubic meters per hour, including 180 outdoor and 540 recirculated. Same volume basis; no other air paths in this calculation. No adequacy or safety judgment is requested.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

0.75 / 25 / three / 50

01 **Coordinator:** Total-supply air changes per hour are [1] _____ .

02 **Specialist:** Outdoor-air changes per hour are [2] _____ .

03 **Coordinator:** Outdoor air forms [3] _____ percent of the supply.

04 **Specialist:** Outdoor flow is [4] _____ liters per second.

CHECK THE GAPS

1. three / **2.** 0.75 / **3.** 25 / **4.** 50

Why these answers fit, page 166

Keep the distinction audible

Explain an air-balance report using consistent volume, time, and outdoor-air quantities.

Language for the next case

Use the right saturation reference

dew point

The supplied vapor-side dew point is 5 C; superheat is referenced to saturated vapor at the matching pressure, not the zero-degree bubble point.

Dev: For this blend's vapor reading, use the dew point of five degrees Celsius. Zero is the bubble point at that pressure, not the correct reference for this superheat calculation.

bubble point

The supplied liquid-side bubble point is 40 C; it is the saturated-liquid reference for subcooling at that reading's pressure.

Dev: Use the bubble point of forty degrees Celsius. The liquid calculation is referenced to saturated liquid at its corresponding pressure, not to the dew value.

temperature glide

A blend with temperature glide has distinct bubble and dew temperatures at a given pressure; the supplied reference pairs differ by five kelvin.

Dev: The blend has temperature glide. Its bubble and dew temperatures differ at a given pressure. Keep the two reference labels instead of silently averaging them.

superheat

Twelve degrees Celsius vapor temperature minus the five-degree dew point gives a seven-kelvin temperature difference; the original temperature reading remains twelve Celsius.

Dev: Yes. The superheat is seven kelvin, a temperature difference. We aren't changing the measured temperature or saying the vapor is at seven degrees Celsius.

subcooling

Forty degrees Celsius bubble point minus thirty-five degrees Celsius liquid temperature gives five kelvin subcooling; using dew would incorrectly give ten.

Dev: Correct. The subcooling here is five kelvin. Don't swap the reference merely because the larger subtraction looks like a more substantial result.

matching pressure

The vapor and liquid readings have different pressure-specific reference pairs; a reference from the other location cannot be substituted into the calculation.

Dev: No. Each comparison needs the matching pressure reference for that reading and location. The report already supplies those pairings; crossing between them would invalidate the result.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

Use the right saturation reference

Fictional completed report for a refrigerant blend: at the vapor-reading pressure, bubble point 0 C, dew point 5 C, vapor temperature 12 C; at the liquid-reading pressure, bubble point 40 C, dew point 45 C, liquid temperature 35 C. Each reference matches its reading's pressure. Superheat = vapor minus dew; subcooling = bubble minus liquid. No targets, diagnosis, or servicing instructions are supplied.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

temperature glide / superheat / subcooling / dew point / bubble point / matching pressure

01 **Sara:** Dev, the completed report gives two saturation temperatures for each pressure. I used zero for the vapor calculation and got twelve degrees of superheat.

02 **Dev:** For this blend's vapor reading, use the [1] _____ of five degrees Celsius. Zero is the bubble point at that pressure, not the correct reference for this superheat calculation.

03 **Sara:** Then twelve minus five gives seven. I need to correct the derived result, but preserve twelve as the original vapor temperature.

04 **Dev:** Yes. The [2] _____ is seven kelvin, a temperature difference. We aren't changing the measured temperature or saying the vapor is at seven degrees Celsius.

05 **Sara:** The liquid-side entry gives forty bubble and forty-five dew, with liquid at thirty-five. Which reference belongs with that calculation?

06 **Dev:** Use the [3] _____ of forty degrees Celsius. The liquid calculation is referenced to saturated liquid at its corresponding pressure, not to the dew value.

Conversation / continued

07 **Sara:** Forty minus thirty-five is five. Using forty-five would give ten and overstate the result by another five.

08 **Dev:** Correct. The [4] _____ here is five kelvin. Don't swap the reference merely because the larger subtraction looks like a more substantial result.

09 **Sara:** Why aren't the bubble and dew points identical? I expected one saturation temperature, as though the whole change of state happened at a single value.

10 **Dev:** The blend has [5] _____. Its bubble and dew temperatures differ at a given pressure. Keep the two reference labels instead of silently averaging them.

Conversation continues on the next page.

Use the right saturation reference

CONVERSATION / CONTINUED

11 **Sara:** Both pairs in this example differ by five. Can I use the vapor-side dew point with the liquid temperature just because both numbers appear in one report?

12 **Dev:** No. Each comparison needs the [6] _____ reference for that reading and location. The report already supplies those pairings; crossing between them would invalidate the result.

13 **Sara:** Is seven kelvin a different-sized interval from seven Celsius degrees? I can see how the report's units might confuse someone reading quickly.

14 **Dev:** Those intervals are the same size. Kelvin is useful for stating a temperature difference here; it doesn't turn the seven-kelvin difference into an absolute temperature of seven kelvin.

15 **Sara:** Do seven and five tell us the system has the correct charge or that an expansion device needs adjustment? The summary asks for the service outcome.

16 **Dev:** Not from these figures alone. No applicable targets or full assessment are supplied. Correct arithmetic doesn't establish system condition, refrigerant quantity, or a remedy.

17 **Sara:** Then I'll make the handoff clear: we've corrected the calculations, not completed servicing. We haven't obtained new readings or changed a setting.

18 **Dev:** Exactly. We're reviewing a completed record. Actual testing, refrigerant work, and adjustments require the relevant qualified procedures and authorization, not a language-book calculation.

19 **Sara:** I'll report vapor twelve Celsius against dew five: superheat seven kelvin. Liquid thirty-five against bubble forty: subcooling five kelvin, with the original location references retained.

20 **Dev:** That's clear. Keep the underlying readings, pressure-specific references, and unresolved assessment status visible so the corrected summary doesn't become an unsupported all-clear.

CHECK THE GAPS

1. dew point / **2.** superheat / **3.** bubble point / **4.** subcooling / **5.** temperature glide / **6.** matching pressure

Why these answers fit, page 168

Recalculate without changing the readings

Completed fictional blend record: vapor temperature 11 C with matching dew 4 C and bubble -1 C; liquid temperature 31 C with matching bubble 37 C and dew 42 C. Use the correct reference for each calculation. No operating target or diagnosis is supplied.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

seven / six / dew / bubble

01 **Coordinator:** The vapor calculation uses the [1] _____ point.

02 **Technician:** The superheat is [2] _____ kelvin.

03 **Coordinator:** The liquid calculation uses the [3] _____ point.

04 **Technician:** The subcooling is [4] _____ kelvin.

CHECK THE GAPS

1. dew / **2.** seven / **3.** bubble / **4.** six

Why these answers fit, page 168

Keep the distinction audible

Explain superheat and subcooling in a completed report without using the wrong blend reference.

01 / Answers and explanations

A / Read the situation

A1. What does 22 degrees Celsius represent?

B. The thermostat setpoint

The brief identifies the number as the target setting rather than a measured room condition.

Return to the exercise, page 4

A2. When do staff report feeling warm?

C. After 14:00

The supplied pattern is an afternoon report beginning after 14:00.

Return to the exercise, page 4

A3. What is established about the equipment?

B. No cause or fault has been established

The complaint and setpoint do not establish a cause or equipment condition.

Return to the exercise, page 4

D / Response choices

D1. Which wording accurately uses the number?

A. The thermostat setpoint is 22 degrees Celsius

The value identifies a target setting, not any of the proposed measurements.

Return to the exercise, page 8

D2. Which summary preserves the complaint?

B. Staff feel warm in the meeting room after 14:00; ambient reading not supplied

The summary keeps attribution, place, period, and missing measurement intact.

Return to the exercise, page 8

01 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Temperature of the surrounding air at the relevant location. Which term fits?

C. ambient temperature

A Not here. latent cooling: Heat removal associated here with condensing water vapor from air. The usual combination is "account for latent cooling".

B Not here. equipment fault: Established malfunction of a system component. The usual combination is "avoid assuming an equipment fault".

C Correct. ambient temperature: Temperature of the surrounding air at the relevant location. The usual combination is "record ambient temperature".

D Not here. sensible cooling: Heat removal associated with lowering air temperature without counting moisture condensation. The usual combination is "distinguish sensible cooling".

Return to the exercise, page 8

D4. Rebuild the exchange from "Twenty-two is the target, not the finding". Asha: "The reports concern the meeting room after two in the afternoon. I don't have the same complaint about every room or every part of the day." Ben: [missing response] Asha: "Will the service team dismiss the complaint if the request says the thermostat is set to twenty-two? The people still feel uncomfortable." Which response belongs in the gap?

D. That is a useful afternoon pattern. I will retain the meeting-room location and after 14:00, so the request does not become a whole-office complaint.

A Not here. This line belongs to turn 12, after Asha says: "Please include Celsius. Our overseas colleagues sometimes use Fahrenheit, and I'd rather the unit didn't disappear when the request is forwarded." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Asha says: "I had written possible sensor failure to make the request more specific. But the information doesn't establish a failed sensor, does it?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Asha says: "Ben, the meeting room is twenty-two degrees, but staff keep saying it feels warm after lunch. How should I describe that without sounding contradictory?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Ben's response in turn 6. It connects Asha's preceding point with Asha's follow-up. Afternoon pattern preserves the reported period after 14:00 without inventing continuous discomfort or exact onset.

Return to the exercise, page 8

E / Conversation

1. setpoint

Turn 2. Setpoint identifies the target setting, which must not be mistaken for a room measurement.

01 / Answers and explanations / continued

2. ambient temperature

Turn 4. Ambient temperature is the surrounding-air condition, for which no confirmed reading is supplied.

3. afternoon pattern

Turn 6. Afternoon pattern preserves the reported period after 14:00 without inventing continuous discomfort or exact onset.

4. occupant feedback

Turn 8. Occupant feedback records the staff experience alongside the target rather than treating one as a contradiction.

5. equipment fault

Turn 10. Equipment fault has not been established by the setpoint or the comfort report.

6. degrees Celsius

Turn 12. Degrees Celsius supplies the unit attached to the stated target value.

7. temperature reading

Turn 14. Temperature reading is absent from this account; that does not assert that no measurement ever exists.

8. service description

Turn 16. Service description communicates the known complaint without prescribing tests or a control adjustment.

9. attributed report

Turn 18. Attributed report links the comfort account to staff and keeps the target and missing reading separate.

10. target temperature

Turn 20. Target temperature is the intended control value, not proof of actual room conditions or performance.

Return to the conversation, page 10

01 / Answers and explanations / continued

F / Accuracy and scope

F1. What should Ben avoid inferring?

C. A particular equipment fault

No diagnostic evidence establishes a component failure in the supplied facts.

Return to the exercise, page 12

F2. Which question best clarifies the displayed value?

A. Does the record label 22 as the target or the measured room temperature?

The bounded question identifies the value using existing information without assuming a fault.

Return to the exercise, page 12

F / Guided speaking

1. setpoint

Manager: The thermostat setpoint is 22 degrees Celsius.

Turn 1. Setpoint correctly identifies the target rather than a measured ambient temperature.

2. 14:00

Coordinator: Staff feel warm after 14:00.

Turn 2. The time preserves the supplied afternoon complaint without inventing another period.

3. ambient

Manager: No confirmed ambient reading is supplied.

Turn 3. Ambient identifies the actual surrounding-air measurement that is missing from the account.

4. unestablished

Coordinator: The cause remains unestablished.

Turn 4. Neither the setpoint nor the report establishes a technical cause.

Return to the conversation, page 13

01 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Twenty-two is the target, not the finding". Asha: "When we say no confirmed reading supplied, we mean this account lacks it. We don't know whether someone has measured the room on another occasion." Ben: [missing response] Asha: "I'll keep the request to the existing information. I don't want my summary mistaken for an instruction to change the thermostat or test a component." Which response belongs in the gap?

B. Yes. The missing temperature reading is a limit of this information. It does not establish the history of every measurement or the condition of the equipment.

A Not here. This line belongs to turn 20, after Asha says: "That preserves the staff account without inventing a measured temperature above target. I'll stop describing twenty-two as the actual room condition." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Ben's response in turn 14. It connects Asha's preceding point with Asha's follow-up. Temperature reading is absent from this account; that does not assert that no measurement ever exists.

C Not here. This line belongs to turn 18, after Asha says: "Could you read it back with the target and missing measurement separate? I need a short version for the facilities manager." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Asha says: "I had written possible sensor failure to make the request more specific. But the information doesn't establish a failed sensor, does it?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 13

02 / Answers and explanations

A / Read the situation

A1. Which label appears in the service request?

C. AC-4

AC-4 is the request label, distinct from the plan and register references.

Return to the exercise, page 14

A2. What does the register identify as AH-4?

A. An indoor air handler

The indoor equipment register explicitly identifies an air handler as AH-4.

Return to the exercise, page 14

A3. What is unresolved?

D. The relationship between CU-4 and AH-4

The supplied documents do not confirm how the two components are related.

Return to the exercise, page 14

D / Response choices

D1. Which summary is accurate?

B. Request AC-4; roof plan CU-4; register AH-4; relationship unverified

The summary preserves the separate source labels and unresolved relationship.

Return to the exercise, page 18

D2. What should Owen ask Mira to provide?

D. The relevant existing register information for reconciliation

Existing source records support reconciliation; renaming, pairing by number, or borrowing an unrelated room would add unsupported information.

Return to the exercise, page 18

02 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Label identifying a particular part of a system. Which term fits?

C. component reference

A Not here. unverified link: Relationship that has not been confirmed. The usual combination is "retain an unverified link".

B Not here. equipment pairing: Confirmed relationship between components intended to operate together. The usual combination is "verify the equipment pairing".

C Correct. component reference: Label identifying a particular part of a system. The usual combination is "retain the component reference".

D Not here. model number: Manufacturer identifier for a product design or version. The usual combination is "confirm the model number".

Return to the exercise, page 18

D4. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "They all end in four. I thought that would be enough to link the indoor and outdoor entries, but the records don't actually confirm it." Owen: [missing response] Mira: "I have the register here already. Can we resolve the question through the records process instead of asking someone to go onto the roof?" Which response belongs in the gap?

A. Then leave the equipment pairing unverified. The shared numeral suggests a question to resolve; it doesn't establish that those components operate together.

A Correct. This is Owen's response in turn 6. It connects Mira's preceding point with Mira's follow-up. Equipment pairing is the relationship that remains unverified despite the shared numeral.

B Not here. This line belongs to turn 8, after Mira says: "I have the register here already. Can we resolve the question through the records process instead of asking someone to go onto the roof?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 18, after Mira says: "Please phrase the query without saying one document must be wrong. The labels may describe different things correctly, even if their connection is missing." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Mira says: "Owen, the school's request says AC-4, but I can't match that cleanly to the project documents. I don't want to send the wrong reference." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 18

E / Conversation

1. record source

Turn 2. Record source identifies where each label comes from instead of treating the labels as interchangeable.

02 / Answers and explanations / continued

2. air handler

Turn 4. Air handler is the indoor item named AH-4 in the register, not a confirmed synonym for CU-4.

3. equipment pairing

Turn 6. Equipment pairing is the relationship that remains unverified despite the shared numeral.

4. equipment register

Turn 8. Equipment register is already available to Mira and can support the document reconciliation.

5. service request

Turn 10. Service request retains the original AC-4 wording while its relationship to other labels is clarified.

6. system reference

Turn 12. System reference may identify the overall arrangement rather than one physical component.

7. model number

Turn 14. Model number identifies a product design or version, unlike the individual-unit serial number.

8. served area

Turn 16. Served area must be established for the identified system rather than borrowed from another request.

9. cross-reference

Turn 18. Cross-reference asks how the documented labels correspond without presuming that any one record is wrong.

10. confirmed mapping

Turn 20. Confirmed mapping supplies verified correspondence before labels are merged or the request is relabeled.

Return to the conversation, page 20

F / Accuracy and scope

F1. Why is the numeral four insufficient?

B. Similar numbering does not establish verified component relationships

An identifier pattern is not evidence that the recorded components form a confirmed pairing.

Return to the exercise, page 22

02 / Answers and explanations / continued

F2. What should remain visible until clarified?

A. The source of each label and the unverified link

Keeping sources and uncertainty supports accurate reconciliation without losing original information.

Return to the exercise, page 22

F / Guided speaking

1. AC-4

Manager: The request says AC-4.

Turn 1. AC-4 is the original request label and must remain traceable.

2. CU-4

Coordinator: The roof plan identifies the outdoor unit as CU-4.

Turn 2. CU-4 belongs to the outdoor item on the roof plan.

3. AH-4

Manager: The register identifies the air handler as AH-4.

Turn 3. AH-4 is the indoor air-handler reference in the register.

4. unverified

Coordinator: Their relationship remains unverified.

Turn 4. Similar numbering does not establish a confirmed link between the recorded components.

Return to the conversation, page 23

02 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "There may be model and serial fields in the register too. Those aren't interchangeable with the school's asset tags, are they?" Owen: [missing response] Mira: "Another request mentions the meeting room. Can I borrow that location to make this one more useful, or would that introduce an unsupported connection?" Which response belongs in the gap?

D. No. A model number identifies a product design or version; a serial number distinguishes a manufactured unit. Keep each field with its recorded equipment item.

A Not here. This line belongs to turn 10, after Mira says: "Would it help if I changed AC-4 to CU-4 on the request now? At least that label would agree with the roof plan." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Mira says: "Owen, the school's request says AC-4, but I can't match that cleanly to the project documents. I don't want to send the wrong reference." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Mira says: "Another request mentions the meeting room. Can I borrow that location to make this one more useful, or would that introduce an unsupported connection?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Owen's response in turn 14. It connects Mira's preceding point with Mira's follow-up. Model number identifies a product design or version, unlike the individual-unit serial number.

Return to the exercise, page 23

03 / Answers and explanations

A / Read the situation

A1. Which opening does the drawing identify as supply?

B. S3

The current drawing explicitly labels S3 as supply air.

Return to the exercise, page 26

A2. Where does Tomas report discomfort?

B. Below S3

The client names the area below S3 as the complaint location.

Return to the exercise, page 26

A3. What is missing?

C. Airflow measurements

No measured airflow data are supplied with the discomfort report.

Return to the exercise, page 26

D / Response choices

D1. Which explanation is accurate?

B. Supply delivers air into the room; return carries air back toward the system

The directional distinction explains the stated functions without changing them into performance findings.

Return to the exercise, page 30

D2. What can the drawing labels establish here?

C. The stated functions of S3 and R2

The supplied drawing labels identify functions, not measured operating conditions.

Return to the exercise, page 30

03 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Grille associated with air returning from the space. Which term fits?

C. return grille

A Not here. draft: Air movement perceived as unwanted by an occupant. The usual combination is "describe a reported draft".

B Not here. air distribution: Delivery and movement of air within a served area. The usual combination is "describe air distribution".

C Correct. return grille: Grille associated with air returning from the space. The usual combination is "identify the return grille".

D Not here. design airflow: Airflow value specified by the design. The usual combination is "compare design airflow".

Return to the exercise, page 30

D4. Rebuild the exchange from "Two ceiling openings, two functions". Tomas: "Then the complaint belongs below the supply opening, S3. Please keep that location rather than writing that the whole room's ventilation has failed." Leila: [missing response] Tomas: "I was about to say S3 delivers too much air. That's how it feels here, but I don't have a measured figure to attach." Which response belongs in the gap?

A. I'll retain it in the distribution report: discomfort below S3. We can identify the drawing label without claiming that it verifies how the installed system performs.

A Correct. This is Leila's response in turn 6. It connects Tomas's preceding point with Tomas's follow-up. Distribution report records the air-related complaint and its location without adding measurements.

B Not here. This line belongs to turn 10, after Tomas says: "An earlier report used CFM. I assumed that was how fast the air moves. Does it actually describe something different?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Tomas says: "They look like parts of the same system. Does R2 also send air into the room, just from a different position?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 12, after Tomas says: "So a feeling of air movement at my seat doesn't give us a CFM figure. And liters per second would be another volume-flow unit." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 30

E / Conversation

1. supply air

Turn 2. Supply air is the stated delivery function of S3 on the current drawing.

03 / Answers and explanations / continued

2. return air

Turn 4. Return air describes the flow back toward the system rather than delivery into the room.

3. distribution report

Turn 6. Distribution report records the air-related complaint and its location without adding measurements.

4. measured airflow

Turn 8. Measured airflow would require relevant measurement data, which Tomas has not supplied.

5. cubic feet per minute

Turn 10. Cubic feet per minute is a volume-flow unit, not an air-speed unit.

6. Air velocity

Turn 12. Air velocity is the speed of air movement and must not be confused with volume flow.

7. balancing

Turn 14. Balancing involves technical distribution work and verification, not merely identifying supply and return labels.

8. damper

Turn 16. Damper is an airflow-control component; no adjustment to it is supported by this conversation.

9. occupied zone

Turn 18. Occupied zone is the area used by people, helping locate the comfort issue rather than diagnose equipment.

10. performance finding

Turn 20. Performance finding would require supporting evidence beyond the labels and discomfort account.

Return to the conversation, page 32

F / Accuracy and scope

F1. Which report invents a measurement?

D. S3 delivers too many liters per second

No airflow measurements support a quantified excess at S3.

Return to the exercise, page 34

03 / Answers and explanations / continued

F2. What is the proper scope of Leila's response?

D. Explain labels and preserve the complaint for assessment

The case supports explanation and accurate reporting, not adjustment or performance conclusions.

Return to the exercise, page 34

F / Guided speaking

1. supply

Technician: S3 is labeled supply air.

Turn 1. Supply is the stated function assigned to S3 by the drawing.

2. return

Client: R2 is labeled return air.

Turn 2. Return distinguishes R2 from the supply opening named in the complaint.

3. S3

Technician: The discomfort is reported below S3.

Turn 3. S3 is the specific location identified by the client.

4. measurements

Client: No airflow measurements have been supplied.

Turn 4. The complaint includes no measured airflow data to support a performance conclusion.

Return to the conversation, page 35

03 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Two ceiling openings, two functions". Tomas: "Since we know which opening is the return, can we at least say the system is balanced? The drawing appears to show a complete arrangement." Leila: [missing response] Tomas: "Someone suggested closing a grille to stop the discomfort. I don't want the service request to present that suggestion as the approved remedy." Which response belongs in the gap?

C. Identifying the openings doesn't establish balancing. That involves technical distribution work and verification, not simply checking that supply and return labels exist.

A Not here. This line belongs to turn 2, after Tomas says: "Leila, the ceiling outlet above this seat is uncomfortable to sit under. It's S3 on the drawing. The other outlet is R2." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Tomas says: "I was about to say S3 delivers too much air. That's how it feels here, but I don't have a measured figure to attach." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Leila's response in turn 14. It connects Tomas's preceding point with Tomas's follow-up. Balancing involves technical distribution work and verification, not merely identifying supply and return labels.

D Not here. This line belongs to turn 18, after Tomas says: "Please include that it's the seat below S3 that concerns me. Saying air-conditioning problem leaves the person reading it with very little to locate." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 35

04 / Answers and explanations

A / Read the situation

A1. What does proposal A's 5 kilowatts describe?

C. Cooling capacity

Proposal A explicitly labels the value as cooling capacity.

Return to the exercise, page 36

A2. What does proposal B's 5 kilowatts describe?

D. Electrical input

Proposal B labels the value electrical input, which is a different quantity.

Return to the exercise, page 36

A3. What conclusion is supported?

C. The numbers alone cannot establish equivalent systems

The matching unit and value do not make the two differently labeled quantities equivalent.

Return to the exercise, page 36

D / Response choices

D1. Which comparison is valid from the supplied entries?

C. A's value is cooling capacity; B's is electrical input

Only the differently labeled quantities can be identified; performance equivalence is not established.

Return to the exercise, page 41

D2. What additional information is needed?

C. Full data sheets, rating conditions, and component pairings

Matching quantities, conditions, and actual equipment combinations are needed; headline numbers or refrigerant names do not fill those gaps.

Return to the exercise, page 41

04 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Unit of energy representing one kilowatt over one hour. Which term fits?

C. kilowatt-hour

A Not here. like-for-like comparison: Comparison of matching quantities on a consistent basis. The usual combination is "make a like-for-like comparison".

B Not here. refrigerant charge: Quantity of refrigerant specified or present in a system. The usual combination is "distinguish refrigerant charge".

C Correct. kilowatt-hour: Unit of energy representing one kilowatt over one hour. The usual combination is "distinguish a kilowatt-hour".

D Not here. nominal capacity: Stated classification or approximate rating rather than every operating output. The usual combination is "distinguish nominal capacity".

Return to the exercise, page 41

D4. Rebuild the exchange from "Five kilowatts of what?". Jules: "Could I divide A's five by B's five and report a performance ratio of one? They're both power figures, after all." Farah: [missing response] Jules: "Then asking each supplier to confirm five kilowatts again wouldn't fix it. We need both quantity labels and the missing values for each proposal." Which response belongs in the gap?

B. Not for an efficiency comparison. You would be combining one system's output with another system's input, rather than corresponding quantities for each proposed arrangement.

A Not here. This line belongs to turn 4, after Jules says: "B says electrical input. I treated kilowatts as though they always told us how much heat the system could remove from the room." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Farah's response in turn 6. It connects Jules's preceding point with Jules's follow-up. Efficiency comparison needs corresponding output and input information on a consistent basis.

C Not here. This line belongs to turn 8, after Jules says: "Then asking each supplier to confirm five kilowatts again wouldn't fix it. We need both quantity labels and the missing values for each proposal." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Jules says: "I'll remove equal-size and ask for complete data sheets, actual pairings, and both output and input values with their conditions." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 41

E / Conversation

1. cooling capacity

Turn 2. Cooling capacity identifies A's heat-removal output rating, not its electrical power input.

04 / Answers and explanations / continued

2. Electrical input

Turn 4. Electrical input is the power required by the equipment, not the useful cooling output.

3. efficiency

Turn 6. Efficiency comparison needs corresponding output and input information on a consistent basis.

4. data sheet

Turn 8. Data sheet supplies product characteristics beyond the two incomplete proposal entries.

5. equipment combination

Turn 10. Equipment combination identifies the components whose joint specifications are relevant to the proposal.

6. refrigerant designation

Turn 12. Refrigerant designation identifies the specified fluid but does not establish system equivalence or replacement approval.

7. substitution review

Turn 14. Substitution review evaluates a proposed replacement and is not bypassed by a familiar refrigerant name.

8. Refrigerant charge

Turn 16. Refrigerant charge concerns the quantity of working fluid, distinct from capacity and refrigerant identity.

9. like-for-like comparison

Turn 18. Like-for-like comparison uses matching quantities and conditions instead of the superficially matching five-kilowatt figures.

10. selection basis

Turn 20. Selection basis is the supporting information still needed before an equipment choice can be approved.

Return to the conversation, page 43

04 / Answers and explanations / continued

F / Accuracy and scope

F1. Which conclusion exceeds the evidence?

A. The systems have equal efficiency

The two entries do not provide the corresponding output and input data needed for that claim.

Return to the exercise, page 46

F2. What does a refrigerant designation provide?

C. Identification to be checked against the actual equipment specification

The designation identifies the fluid but does not establish compatibility or authorization.

Return to the exercise, page 46

F / Guided speaking

1. capacity

Client: A gives 5 kilowatts of cooling capacity.

Turn 1. Capacity identifies the cooling-output quantity stated in proposal A.

2. input

Estimator: B gives 5 kilowatts of electrical input.

Turn 2. Input identifies power required by the equipment, not cooling output.

3. sheets

Client: We need complete data sheets and pairings.

Turn 3. Full data sheets and component pairings are missing from the supplied proposals.

4. approved

Estimator: No selection is yet approved.

Turn 4. The comparison does not establish an accepted equipment choice or authorization.

Return to the conversation, page 47

04 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Five kilowatts of what?". Jules: "I don't want the supplier to read our question as permission to use another fluid. We're asking for information, not changing the specified equipment." Farah: [missing response] Jules: "The sheet also has a refrigerant amount. Is that a different statement of capacity, or is it simply another product detail?" Which response belongs in the gap?

C. Then keep any substitution review separate and explicitly pending. A familiar refrigerant name doesn't bypass that process or authorize refrigerant handling.

A Not here. This line belongs to turn 8, after Jules says: "Then asking each supplier to confirm five kilowatts again wouldn't fix it. We need both quantity labels and the missing values for each proposal." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Jules says: "The sheet also has a refrigerant amount. Is that a different statement of capacity, or is it simply another product detail?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Farah's response in turn 14. It connects Jules's preceding point with Jules's follow-up. Substitution review evaluates a proposed replacement and is not bypassed by a familiar refrigerant name.

D Not here. This line belongs to turn 4, after Jules says: "B says electrical input. I treated kilowatts as though they always told us how much heat the system could remove from the room." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 47

05 / Answers and explanations

A / Read the situation

A1. What did Elena report yesterday?

D. Water near AH-6 at 15:00

Elena's account concerns observed water near AH-6 at the stated time.

Return to the exercise, page 50

A2. What does today's visit note establish?

D. The nearby floor was dry and the source was not identified

The note records a later surface condition and an unresolved source.

Return to the exercise, page 50

A3. What does the earlier drain-pan entry prove?

A. Only that a concern was recorded, not a causal link

The historical concern is not linked to the water by the supplied evidence.

Return to the exercise, page 50

D / Response choices

D1. Which summary preserves chronology?

D. Water near AH-6 yesterday at 15:00; floor dry today; source unidentified

The accurate summary retains near AH-6, yesterday's 15:00 report, today's dry floor, and the unidentified source without changing attribution or inventing cause.

Return to the exercise, page 54

D2. Which wording improperly asserts origin?

C. Water came from AH-6

The supplied report gives proximity, not proof that AH-6 produced the water.

Return to the exercise, page 54

05 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Account describing water or dampness at a location. Which term fits?

A. moisture report

A Correct. moisture report: Account describing water or dampness at a location. The usual combination is "retain the moisture report".

B Not here. finding: Observation or conclusion established through the relevant assessment. The usual combination is "state the actual finding".

C Not here. reported water: Water described in an account before its cause is established. The usual combination is "locate the reported water".

D Not here. intermittent symptom: Condition that may appear at some times and not others. The usual combination is "describe an intermittent symptom".

Return to the exercise, page 54

D4. Rebuild the exchange from "Yesterday wet, today dry, source still unknown". Elena: "The earlier service log mentions a drain-pan concern. Someone suggested putting that as the cause so the record has an explanation." Kai: [missing response] Elena: "I only know the water was near the air handler. Writing that it came from AH-6 would be more specific than what I observed." Which response belongs in the gap?

A. We should retain the drain-pan concern as history, but the entry does not establish a connection to yesterday's report. It is not enough to name the pan as the proven source.

A Correct. This is Kai's response in turn 6. It connects Elena's preceding point with Elena's follow-up. Drain-pan concern is historical information, not a confirmed cause of the later water report.

B Not here. This line belongs to turn 18, after Elena says: "Please combine the three records without losing their sources: my account, today's visit note, and the earlier drain-pan entry." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 10, after Elena says: "The old wording is concern, not failure. Could you retain that distinction rather than turning a historical note into a confirmed broken part?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Elena says: "That's the chronology I need. It neither dismisses yesterday's observation nor treats today's dry floor as proof that the source has been fixed." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 54

E / Conversation

1. chronology

Turn 2. Chronology keeps yesterday's report and today's observation separate instead of treating them as simultaneous claims.

05 / Answers and explanations / continued

2. visit note

Turn 4. Visit note is the source for today's dry-floor observation, not yesterday's client report.

3. drain-pan concern

Turn 6. Drain-pan concern is historical information, not a confirmed cause of the later water report.

4. evidence boundary

Turn 8. Evidence boundary limits the account to observed location rather than asserting origin from AH-6.

5. historical entry

Turn 10. Historical entry must retain its recorded status as a concern rather than being upgraded to confirmed failure.

6. source identification

Turn 12. Source identification remains unresolved because today's note explicitly says the source was not identified.

7. repair status

Turn 14. Repair status cannot be inferred from a later dry surface when no corrective work is documented.

8. recurrence history

Turn 16. Recurrence history requires evidence of repeated events, not merely an earlier concern about a component.

9. causal link

Turn 18. Causal link is the missing connection between the earlier concern and the reported water.

10. record reconciliation

Turn 20. Record reconciliation combines the three sources while preserving time, attribution, and uncertainty.

Return to the conversation, page 56

05 / Answers and explanations / continued

F / Accuracy and scope

F1. How should the drain-pan entry be handled?

C. Retain it as history with no established causal link

Relevant history can be preserved without asserting a cause or resolution.

Return to the exercise, page 58

F2. What can the dry-floor note establish about repair?

B. No completed repair by itself

A surface observation is not evidence that corrective work occurred.

Return to the exercise, page 58

F / Guided speaking

1. 15:00

Client: Water was reported near AH-6 at 15:00 yesterday.

Turn 1. The time belongs to Elena's report on the previous day.

2. dry

Coordinator: Today the nearby floor was recorded as dry.

Turn 2. Dry describes the surface observed during the later visit.

3. concern

Client: The earlier log mentions a drain-pan concern.

Turn 3. Concern preserves the historical wording without upgrading it to a confirmed failure.

4. identified

Coordinator: The source was not identified.

Turn 4. The visit note leaves the origin of the reported water unresolved.

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05 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Yesterday wet, today dry, source still unknown". Elena: "Does dry today imply someone repaired it overnight? I haven't supplied any record of work between my report and the visit." Kai: [missing response] Elena: "One older concern also doesn't establish a pattern of repeated water events. We shouldn't attach a frequency to the problem without the history." Which response belongs in the gap?

D. No. The repair status cannot be inferred from the dry-floor observation. Nothing in these records establishes that a repair was completed between your report and the visit.

A Not here. This line belongs to turn 4, after Elena says: "Please keep yesterday and fifteen hundred attached to my account. Otherwise it could sound as though the technician saw water on today's visit." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Elena says: "One older concern also doesn't establish a pattern of repeated water events. We shouldn't attach a frequency to the problem without the history." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Elena says: "Kai, yesterday I reported water near AH-6 at three. Today's note says dry floor. I don't want my report dropped because the visit happened later." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Kai's response in turn 14. It connects Elena's preceding point with Elena's follow-up. Repair status cannot be inferred from a later dry surface when no corrective work is documented.

Return to the exercise, page 59

06 / Answers and explanations

A / Read the situation

A1. What was recorded at 06:10?

C. A cabinet display value of 8 degrees Celsius

The record supplies a cabinet display value, not product temperatures or duration.

Return to the exercise, page 60

A2. What do the matching timestamps establish?

B. The two entries share a recorded time

Coincident timestamps do not establish cause, duration, or product safety.

Return to the exercise, page 60

A3. Who must decide the food-safety question?

A. The responsible food-safety lead under site procedures

The scenario assigns that decision to the responsible lead using the site's procedures.

Return to the exercise, page 60

D / Response choices

D1. Which summary belongs in the escalation?

C. Display 8 degrees Celsius at 06:10; product data absent; cause and duration unconfirmed

The summary preserves the actual value and all material information limits.

Return to the exercise, page 64

D2. Why is defrost caused it unsupported?

D. The shared timestamp does not establish causation

Coincident entries show a shared recorded time, not a verified cause, completed cycle, duration, or food-temperature history.

Return to the exercise, page 64

06 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Number shown by the equipment interface. Which term fits?

B. display value

A Not here. event duration: Length of time an event continued. The usual combination is "establish event duration".

B Correct. display value: Number shown by the equipment interface. The usual combination is "record the display value".

C Not here. food-safety lead: Person assigned responsibility for the relevant food-safety decision. The usual combination is "notify the food-safety lead".

D Not here. trend record: Series of values showing change over time. The usual combination is "review the trend record".

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D4. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "The alarm history and the scheduled-defrost entry both say 06:10. Could I describe the alarm as caused by defrost?" Imani: [missing response] Noah: "The owner asked how long it lasted. There's only that one recorded time in this extract, with no confirmed ending time." Which response belongs in the gap?

D. The shared timestamp only tells us the entries coincide. It doesn't establish the cause, prove the event was harmless, or determine what happened to the products.

A Not here. This line belongs to turn 12, after Noah says: "Please distinguish the scheduled item from the actual explanation. I want the handoff to say what the log contains without implying we've diagnosed the event." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Noah says: "That's all I've been given for temperature: the cabinet display. There are no food readings in the information I'm sending to the lead." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 18, after Noah says: "My message says display eight Celsius at 06:10; no product readings supplied; alarm and defrost entries coincide; duration and cause unconfirmed." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Imani's response in turn 6. It connects Noah's preceding point with Noah's follow-up. Timestamp establishes the recorded time shared by the entries, not why the alarm occurred.

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E / Conversation

1. display value

Turn 2. Display value is equipment information and does not supply the missing product temperatures or exposure history.

06 / Answers and explanations / continued

2. product temperature

Turn 4. Product temperature concerns the food itself and is absent from the information supplied.

3. timestamp

Turn 6. Timestamp establishes the recorded time shared by the entries, not why the alarm occurred.

4. event duration

Turn 8. Event duration remains unknown because one timestamp does not establish a complete interval.

5. food-safety lead

Turn 10. Food-safety lead is the assigned decision-maker for product safety under the site's procedures.

6. scheduled defrost

Turn 12. Scheduled defrost is a recorded event sharing the time, not a proven explanation for the alarm.

7. Alarm clearance

Turn 14. Alarm clearance concerns equipment alarm status and does not establish product safety or past exposure.

8. exposure history

Turn 16. Exposure history describes conditions experienced by the products over time, which these isolated records do not establish.

9. escalation summary

Turn 18. Escalation summary passes the actual records and missing information to the responsible lead without authorizing release.

10. Product disposition

Turn 20. Product disposition is the responsible decision about the food, distinct from interpreting the equipment records.

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06 / Answers and explanations / continued

F / Accuracy and scope

F1. Who should receive the food-release question?

D. The responsible food-safety lead using site procedures

Product decisions require the assigned responsible process rather than an improvised technical assurance.

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F2. Which statement improperly fills a data gap?

C. The food was warm for only one minute

No event duration or product exposure period is established by the record.

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F / Guided speaking

1. 8

Manager: The display recorded 8 degrees Celsius.

Turn 1. Eight is the cabinet display value, not a verified product temperature.

2. 06:10

Technician: The timestamp was 06:10.

Turn 2. The timestamp records one time and does not establish event duration.

3. supplied

Manager: Product temperatures were not supplied.

Turn 3. The absence of product readings is a material limit on the information.

4. lead

Technician: The responsible food-safety lead must decide under site procedures.

Turn 4. The assigned lead makes the product decision rather than the equipment display alone.

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06 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "Suppose a later update says the alarm has disappeared. Would that alone answer whether the affected stock can be used?" Imani: [missing response] Noah: "So I should leave the earlier product conditions unresolved, rather than assume that a cleared display would make the whole episode irrelevant." Which response belongs in the gap?

C. No. Alarm clearance describes alarm status, not the products' earlier conditions. It doesn't replace the responsible food-safety decision or establish an acceptable exposure history.

A Not here. This line belongs to turn 4, after Noah says: "That's all I've been given for temperature: the cabinet display. There are no food readings in the information I'm sending to the lead." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Noah says: "The owner asked how long it lasted. There's only that one recorded time in this extract, with no confirmed ending time." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Imani's response in turn 14. It connects Noah's preceding point with Noah's follow-up. Alarm clearance concerns equipment alarm status and does not establish product safety or past exposure.

D Not here. This line belongs to turn 2, after Noah says: "Imani, we've alerted our food-safety lead and aren't treating this call as clearance to serve the affected food. Can you help explain the cabinet record?" It addresses that point, not the missing turn in the supplied exchange.

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07 / Answers and explanations

A / Read the situation

A1. Why does Priya want the grille moved?

D. It visually conflicts with a pendant light

The request concerns appearance, not an established system fault.

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A2. Which consequences have been assessed?

A. None of the listed change effects

Ductwork, restoration, access, price, and timing remain unassessed.

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A3. What are the stated choices?

C. Retain the reviewed layout or request a coordinated change proposal

The brief supplies two bounded options without authorizing an immediate alteration.

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D / Response choices

D1. Which response explains the choice accurately?

C. Retain the reviewed layout or request a coordinated proposal with impacts assessed

The response offers the two supplied options and preserves the need to assess consequences.

Return to the exercise, page 77

D2. Which item must not disappear from review?

B. Service access

Service access is explicitly one of the unassessed matters in the brief.

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07 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Drawing of ceiling elements represented as viewed in reflection from below. Which term fits?

C. reflected ceiling plan

A Not here. revised layout: Updated arrangement proposed or accepted after a change. The usual combination is "review the revised layout".

B Not here. rigid duct: Air-conveying duct with a relatively fixed form. The usual combination is "distinguish rigid duct".

C Correct. reflected ceiling plan: Drawing of ceiling elements represented as viewed in reflection from below. The usual combination is "review the reflected ceiling plan".

D Not here. coordination review: Evaluation of the change across affected elements and trades. The usual combination is "arrange a coordination review".

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D4. Rebuild the exchange from "Moving a grille is a coordinated change". Priya: "What would need assessment beyond marking a new position on the ceiling plan? I hadn't thought about the connection above it." Daniel: [missing response] Priya: "Would restoring the old ceiling opening be part of that review too? I don't want a price that leaves the finishing work unexplained." Which response belongs in the gap?

D. The ductwork has not been assessed, nor have ceiling restoration, service access, price, or timing. A revised drawing must be supported by review of those related effects.

A Not here. This line belongs to turn 4, after Priya says: "Yes, it's an appearance issue. I'm not reporting poor cooling or saying the grille has failed; I want the light and grille to sit better together." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Priya says: "Can we also check service access? Moving something to a neater position wouldn't help if it makes the relevant services hard to reach." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Priya says: "Daniel, the supply grille looks awkward beside the pendant light. I'd like it moved. Is the small distance on the ceiling a small change to the work?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Daniel's response in turn 6. It connects Priya's preceding point with Priya's follow-up. Ductwork is one of the hidden systems whose changes have not been assessed for the proposed move.

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E / Conversation

1. design preference

Turn 2. Design preference identifies the appearance concern without claiming a performance fault.

07 / Answers and explanations / continued

2. visual conflict

Turn 4. Visual conflict describes the relationship in appearance between the grille and pendant light.

3. ductwork

Turn 6. Ductwork is one of the hidden systems whose changes have not been assessed for the proposed move.

4. Ceiling restoration

Turn 8. Ceiling restoration concerns reinstating affected finishes and remains part of the unassessed change scope.

5. access provision

Turn 10. Access provision concerns how the relevant services remain reachable after a proposed layout change.

6. reviewed layout

Turn 12. Reviewed layout is the existing arrangement available as the first stated option.

7. change proposal

Turn 14. Change proposal is the requested assessed alternative, not authorization to install or acceptance of its terms.

8. schedule impact

Turn 16. Schedule impact remains unassessed, so unchanged completion cannot be guaranteed.

9. cost impact

Turn 18. Cost impact must be assessed with the defined work rather than assumed absent because the visible move seems small.

10. decision record

Turn 20. Decision record preserves the selected assessment step without turning it into an instruction to proceed.

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07 / Answers and explanations / continued

F / Accuracy and scope

F1. What does requesting a proposal authorize here?

A. Assessment of the change, not its installation

A request for review does not equal agreement to carry out the proposed work.

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F2. Which phrase overstates certainty?

D. We can move it with no cost or timing effect

Cost and timing effects remain unknown, so a no-impact promise is unsupported.

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F / Guided speaking

1. layout

Coordinator: You can retain the reviewed layout.

Turn 1. Layout refers to the existing reviewed arrangement available as the first option.

2. proposal

Client: Or request a coordinated change proposal.

Turn 2. Proposal is the assessed alternative the client may request before deciding.

3. assessment

Coordinator: Cost and timing still need assessment.

Turn 3. The effects on price and schedule have not yet been evaluated.

4. relocation

Client: This request does not authorize relocation.

Turn 4. Requesting a proposal does not approve moving the supply grille.

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07 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Moving a grille is a coordinated change". Priya: "I'd like a proposal. Please don't treat that request as acceptance of an unknown extra cost or as permission to start the move." Daniel: [missing response] Priya: "Can you keep the original finish date? The office move-in depends on it, though I realize the schedule effect hasn't been assessed yet." Which response belongs in the gap?

C. It does not. A change proposal will define the alternative and its consequences for review. Requesting it is separate from accepting the resulting work and commercial terms.

A Not here. This line belongs to turn 4, after Priya says: "Yes, it's an appearance issue. I'm not reporting poor cooling or saying the grille has failed; I want the light and grille to sit better together." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Priya says: "Daniel, the supply grille looks awkward beside the pendant light. I'd like it moved. Is the small distance on the ceiling a small change to the work?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Daniel's response in turn 14. It connects Priya's preceding point with Priya's follow-up. Change proposal is the requested assessed alternative, not authorization to install or acceptance of its terms.

D Not here. This line belongs to turn 6, after Priya says: "What would need assessment beyond marking a new position on the ceiling plan? I hadn't thought about the connection above it." It addresses that point, not the missing turn in the supplied exchange.

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08 / Answers and explanations

A / Read the situation

A1. What does the commissioning report list?

D. A deferred seasonal check

The report explicitly identifies a check that has been deferred rather than completed.

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A2. What responsibility does Nia accept?

B. Document correction and a Friday update

Nia's commitment is to document correction and communication, not the unassigned scheduling task.

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A3. What remains unresolved about the seasonal check?

C. Both the date and the scheduling owner

The check is unbooked and responsibility for arranging it is not confirmed.

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D / Response choices

D1. Which summary preserves all responsibilities?

B. Nia corrects documents and updates Friday; seasonal-check owner and date remain open

The summary keeps Nia's accepted task distinct from the unresolved scheduling responsibility.

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D2. What does Friday identify?

B. Nia's promised update

The only Friday commitment is communication about follow-up.

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08 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Verification associated with relevant seasonal operating conditions. Which term fits?

C. seasonal check

A Not here. operating record: Documented information about actual operation. The usual combination is "retain the operating record".

B Not here. scheduling owner: Person responsible for arranging the required attendance or check. The usual combination is "confirm the scheduling owner".

C Correct. seasonal check: Verification associated with relevant seasonal operating conditions. The usual combination is "schedule the seasonal check".

D Not here. maintenance requirement: Relevant action or condition specified for upkeep. The usual combination is "verify the maintenance requirement".

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D4. Rebuild the exchange from "Report received, seasonal check still open". Ben: "Who is arranging the seasonal check? I also can't find a booked date, so both responsibility and scheduling seem unresolved." Nia: [missing response] Ben: "I nearly assigned that to you because you're handling the wrong sheet. That would add a responsibility you haven't actually accepted." Which response belongs in the gap?

A. The scheduling owner has not been confirmed, and no date is booked. I should not imply that my document-correction task resolves either of those questions.

A Correct. This is Nia's response in turn 6. It connects Ben's preceding point with Ben's follow-up. Scheduling owner has not been confirmed, so the arranging responsibility remains open.

B Not here. This line belongs to turn 18, after Ben says: "Let me check the responsibilities: you own document correction and the Friday update. The seasonal-check owner and date are both still unconfirmed." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Ben says: "I also want the summary to preserve the actual warranty terms, without implying the open check creates a new warranty promise." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Ben says: "Thanks for taking the document correction. I want to separate that from the outstanding check rather than mark everything complete on receipt of the report." It addresses that point, not the missing turn in the supplied exchange.

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E / Conversation

1. document mismatch

Turn 2. Document mismatch identifies the wrong-model maintenance sheet as a correction task.

08 / Answers and explanations / continued

2. deferred check

Turn 4. Deferred check remains outstanding even though its existence is recorded in the delivered report.

3. scheduling owner

Turn 6. Scheduling owner has not been confirmed, so the arranging responsibility remains open.

4. ownership gap

Turn 8. Ownership gap names the unassigned scheduling responsibility rather than silently giving it to Nia.

5. update commitment

Turn 10. Update commitment is Nia's promise to communicate on Friday, not a promised technical-check result.

6. model-specific guidance

Turn 12. Model-specific guidance links the corrected maintenance information to the actual equipment rather than a similar model.

7. performance result

Turn 14. Performance result cannot be generalized to all conditions when a seasonal check is still deferred.

8. warranty terms

Turn 16. Warranty terms retain their actual meaning and are not revised by the document update or deferred-check discussion.

9. responsibility readback

Turn 18. Responsibility readback distinguishes the assigned document task from the unassigned seasonal scheduling task.

10. handover register

Turn 20. Handover register should record receipt alongside corrections and outstanding verification without claiming full completion.

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08 / Answers and explanations / continued

F / Accuracy and scope

F1. Which inference is unsupported?

B. Correcting the maintenance sheet completes the seasonal check

A document correction is not performance of the deferred verification.

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F2. What should the handover keep open?

C. Correct model guidance, deferred verification, scheduling owner, and date

The remaining items are separate; receipt, a document task, and an update commitment do not complete verification, assign scheduling ownership, or book a check.

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F / Guided speaking

1. Nia

Client: Nia owns document correction.

Turn 1. Nia accepts the correction task, not confirmed ownership of arranging the seasonal check.

2. Friday

Coordinator: The update is promised for Friday.

Turn 2. Friday is the communication commitment rather than a booked technical appointment.

3. unbooked

Client: The seasonal check is still unbooked.

Turn 3. No confirmed date has been supplied for the deferred seasonal verification.

4. confirmed

Coordinator: Its scheduling owner is not yet confirmed.

Turn 4. The responsibility for arranging the check remains an explicit open question.

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08 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Report received, seasonal check still open". Ben: "Would report received support a statement that the system has passed every operating condition? The deferred check suggests that would be too broad." Nia: [missing response] Ben: "I also want the summary to preserve the actual warranty terms, without implying the open check creates a new warranty promise." Which response belongs in the gap?

C. No. We have no overall performance result to add from these facts. The deferred item remains part of the record, and we should not invent a successful result for it.

A Not here. This line belongs to turn 12, after Ben says: "Please match the corrected sheet to the actual model. A similar product leaflet would still leave the maintenance information uncertain." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Ben says: "What date can I put down for the next update? I need a communication commitment even while the technical appointment remains unbooked." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Nia's response in turn 14. It connects Ben's preceding point with Ben's follow-up. Performance result cannot be generalized to all conditions when a seasonal check is still deferred.

D Not here. This line belongs to turn 8, after Ben says: "I nearly assigned that to you because you're handling the wrong sheet. That would add a responsibility you haven't actually accepted." It addresses that point, not the missing turn in the supplied exchange.

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Review 1 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Target value used by a control system. Which term fits?

B. setpoint

A Not here. heat gain: Heat added to a space from its surroundings or internal sources. The usual combination is "assess heat gain".

B Correct. setpoint: Target value used by a control system. The usual combination is "distinguish the setpoint".

C Not here. occupant feedback: Account supplied by people using the space. The usual combination is "preserve occupant feedback".

D Not here. target temperature: Temperature the control is intended to seek. The usual combination is "state the target temperature".

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2. A colleague needs the term for this meaning: Equipment assembly located within the building or served space. Which term fits?

B. indoor unit

A Not here. cross-reference: Link connecting corresponding entries in different records. The usual combination is "establish a cross-reference".

B Correct. indoor unit: Equipment assembly located within the building or served space. The usual combination is "identify the indoor unit".

C Not here. service request: Record asking for attention to an identified issue or system. The usual combination is "reconcile the service request".

D Not here. equipment register: Record identifying equipment and relevant asset information. The usual combination is "consult the equipment register".

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Review 1 / Explanations / continued

3. Rebuild the exchange from "Twenty-two is the target, not the finding". Asha: "I had written possible sensor failure to make the request more specific. But the information doesn't establish a failed sensor, does it?" Ben: [missing response] Asha: "Please include Celsius. Our overseas colleagues sometimes use Fahrenheit, and I'd rather the unit didn't disappear when the request is forwarded." Which response belongs in the gap?

A. It is. No equipment fault has been established. We should not name a failed sensor or another component simply to give the complaint a technical-sounding cause.

A Correct. This is Ben's response in turn 10. It connects Asha's preceding point with Asha's follow-up. Equipment fault has not been established by the setpoint or the comfort report.

B Not here. This line belongs to turn 14, after Asha says: "When we say no confirmed reading supplied, we mean this account lacks it. We don't know whether someone has measured the room on another occasion." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 18, after Asha says: "Could you read it back with the target and missing measurement separate? I need a short version for the facilities manager." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Asha says: "That preserves the staff account without inventing a measured temperature above target. I'll stop describing twenty-two as the actual room condition." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 24

Review 1 / Explanations / continued

4. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "Please phrase the query without saying one document must be wrong. The labels may describe different things correctly, even if their connection is missing." Owen: [missing response] Mira: "I'll send all three sources intact. That should let the record owner investigate the actual mapping instead of guessing which label we preferred." Which response belongs in the gap?

D. Ask for a confirmed cross-reference between request AC-4, plan item CU-4, and register item AH-4, including the component relationship and the served area.

A Not here. This line belongs to turn 2, after Mira says: "Owen, the school's request says AC-4, but I can't match that cleanly to the project documents. I don't want to send the wrong reference." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Mira says: "I have the register here already. Can we resolve the question through the records process instead of asking someone to go onto the roof?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 6, after Mira says: "They all end in four. I thought that would be enough to link the indoor and outdoor entries, but the records don't actually confirm it." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Owen's response in turn 18. It connects Mira's preceding point with Mira's follow-up. Cross-reference asks how the documented labels correspond without presuming that any one record is wrong.

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Review 2 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Grille associated with delivery of air into the space. Which term fits?

A. supply grille

A Correct. supply grille: Grille associated with delivery of air into the space. The usual combination is "locate the supply grille".

B Not here. balancing: Technical adjustment and verification of distribution to meet specified requirements. The usual combination is "distinguish balancing from explanation".

C Not here. ductwork: Network of ducts and associated components. The usual combination is "review the ductwork".

D Not here. air changes per hour: Hourly airflow divided by room volume, with the air source specified. The usual combination is "distinguish total and outdoor-air changes".

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2. A colleague needs the term for this meaning: Component arrangement confirmed as suitable together by the relevant specifications. Which term fits?

A. matched system

A Correct. matched system: Component arrangement confirmed as suitable together by the relevant specifications. The usual combination is "identify the matched system".

B Not here. superheat: Vapor temperature above its saturation reference at the same pressure; use dew for the stated blend. The usual combination is "report superheat in kelvin".

C Not here. cooling capacity: Rate at which a system can remove heat under stated conditions. The usual combination is "compare cooling capacity".

D Not here. refrigerant: Working fluid used to transfer heat in a refrigeration system. The usual combination is "identify the specified refrigerant".

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Review 2 / Explanations / continued

3. Rebuild the exchange from "AC-4, CU-4, and AH-4 are not yet a pairing". Mira: "Would it help if I changed AC-4 to CU-4 on the request now? At least that label would agree with the roof plan." Owen: [missing response] Mira: "Right, a request about the whole system might become one about just the outdoor unit. We need the relationship before choosing the final reference." Which response belongs in the gap?

A. Please retain the original service request wording. Add the plan and register references alongside it; changing the label prematurely could narrow what the requester meant.

A Correct. This is Owen's response in turn 10. It connects Mira's preceding point with Mira's follow-up. Service request retains the original AC-4 wording while its relationship to other labels is clarified.

B Not here. This line belongs to turn 16, after Mira says: "Another request mentions the meeting room. Can I borrow that location to make this one more useful, or would that introduce an unsupported connection?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Mira says: "Owen, the school's request says AC-4, but I can't match that cleanly to the project documents. I don't want to send the wrong reference." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 14, after Mira says: "There may be model and serial fields in the register too. Those aren't interchangeable with the school's asset tags, are they?" It addresses that point, not the missing turn in the supplied exchange.

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4. Rebuild the exchange from "Five kilowatts of what?". Jules: "I see four different things now: fluid identity, fluid quantity, output power, and input power. The matching fives only concern two of those." Farah: [missing response] Jules: "I'll remove equal-size and ask for complete data sheets, actual pairings, and both output and input values with their conditions." Which response belongs in the gap?

C. Right. We need a like-for-like comparison using corresponding ratings and conditions. None of the missing values should be filled with the figure from the other proposal.

A Not here. This line belongs to turn 12, after Jules says: "One summary includes a refrigerant name. Could I use that as evidence that the two proposals are interchangeable replacements?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Jules says: "Farah, both proposals show five kilowatts. I put equal-size systems in my comparison sheet and moved straight to the two prices." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Farah's response in turn 18. It connects Jules's preceding point with Jules's follow-up. Like-for-like comparison uses matching quantities and conditions instead of the superficially matching five-kilowatt figures.

D Not here. This line belongs to turn 20, after Jules says: "I'll remove equal-size and ask for complete data sheets, actual pairings, and both output and input values with their conditions." It addresses that point, not the missing turn in the supplied exchange.

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Review 3 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Pipe or tube associated with carrying condensate away. Which term fits?

B. condensate line

A Not here. condensation: Formation of liquid from vapor under relevant conditions. The usual combination is "distinguish condensation from a diagnosis".

B Correct. condensate line: Pipe or tube associated with carrying condensate away. The usual combination is "identify the condensate line".

C Not here. dry observation: Record that the visible surface was dry at the stated time. The usual combination is "qualify the dry observation".

D Not here. chronology: Ordered sequence of events and observations. The usual combination is "preserve the chronology".

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2. A colleague needs the term for this meaning: Defrost event listed in the control schedule or record. Which term fits?

A. scheduled defrost

A Correct. scheduled defrost: Defrost event listed in the control schedule or record. The usual combination is "distinguish scheduled defrost".

B Not here. refrigerated cabinet: Enclosed equipment used to hold products under cooled conditions. The usual combination is "identify the refrigerated cabinet".

C Not here. alarm acknowledgment: Record that an alarm has been noticed or accepted by an operator. The usual combination is "distinguish alarm acknowledgment".

D Not here. exposure history: Record of conditions experienced by the product over time. The usual combination is "establish exposure history".

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Review 3 / Explanations / continued

3. Rebuild the exchange from "Five kilowatts of what?". Jules: "I haven't received the indoor and outdoor pairings. Is a brochure for the outdoor unit enough if the installer chooses the indoor one later?" Farah: [missing response] Jules: "One summary includes a refrigerant name. Could I use that as evidence that the two proposals are interchangeable replacements?" Which response belongs in the gap?

B. No. The equipment combination matters to the stated performance. We need to know which components the proposal actually pairs, not assume any indoor unit gives the same result.

A Not here. This line belongs to turn 18, after Jules says: "I see four different things now: fluid identity, fluid quantity, output power, and input power. The matching fives only concern two of those." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Farah's response in turn 10. It connects Jules's preceding point with Jules's follow-up. Equipment combination identifies the components whose joint specifications are relevant to the proposal.

C Not here. This line belongs to turn 2, after Jules says: "Farah, both proposals show five kilowatts. I put equal-size systems in my comparison sheet and moved straight to the two prices." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 12, after Jules says: "One summary includes a refrigerant name. Could I use that as evidence that the two proposals are interchangeable replacements?" It addresses that point, not the missing turn in the supplied exchange.

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Review 3 / Explanations / continued

4. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "My message says display eight Celsius at 06:10; no product readings supplied; alarm and defrost entries coincide; duration and cause unconfirmed." Imani: [missing response] Noah: "The lead will handle the food decision under our procedures. I'll preserve the technical record and won't describe this call as approval to resume using the stock." Which response belongs in the gap?

B. That's a useful escalation summary. Send any relevant updates through the site process as they become available, keeping the product question separate from the equipment diagnosis.

A Not here. This line belongs to turn 12, after Noah says: "Please distinguish the scheduled item from the actual explanation. I want the handoff to say what the log contains without implying we've diagnosed the event." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Imani's response in turn 18. It connects Noah's preceding point with Noah's follow-up. Escalation summary passes the actual records and missing information to the responsible lead without authorizing release.

C Not here. This line belongs to turn 4, after Noah says: "That's all I've been given for temperature: the cabinet display. There are no food readings in the information I'm sending to the lead." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 6, after Noah says: "The alarm history and the scheduled-defrost entry both say 06:10. Could I describe the alarm as caused by defrost?" It addresses that point, not the missing turn in the supplied exchange.

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Review 4 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Appearance issue between elements in the same view. Which term fits?

D. visual conflict

A Not here. flexible duct: Air-conveying duct with a flexible construction. The usual combination is "identify the flexible duct".

B Not here. change proposal: Defined suggestion for altering the existing arrangement. The usual combination is "request a change proposal".

C Not here. technical feasibility: Whether the proposed arrangement can satisfy relevant technical conditions. The usual combination is "establish technical feasibility".

D Correct. visual conflict: Appearance issue between elements in the same view. The usual combination is "describe the visual conflict".

Return to the exercise, page 96

2. A colleague needs the term for this meaning: Person responsible for resolving a record error. Which term fits?

C. correction owner

A Not here. outstanding verification: Check not yet completed or accepted. The usual combination is "track outstanding verification".

B Not here. commissioning: Process of verifying and documenting system performance against relevant project requirements. The usual combination is "distinguish commissioning from document delivery".

C Correct. correction owner: Person responsible for resolving a record error. The usual combination is "name the correction owner".

D Not here. deferral reason: Explanation for postponing an activity. The usual combination is "preserve the deferral reason".

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Review 4 / Explanations / continued

3. Rebuild the exchange from "An alarm, a display value, and missing product data". Noah: "I'll keep the lead updated with those gaps. Our opening time is close, but the technical call isn't a substitute for the product decision." Imani: [missing response] Noah: "Please distinguish the scheduled item from the actual explanation. I want the handoff to say what the log contains without implying we've diagnosed the event." Which response belongs in the gap?

C. Correct. The food-safety lead must decide under your site procedures. Follow that process while the equipment information is clarified; don't wait for a reassuring guess from the display.

A Not here. This line belongs to turn 6, after Noah says: "The alarm history and the scheduled-defrost entry both say 06:10. Could I describe the alarm as caused by defrost?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 18, after Noah says: "My message says display eight Celsius at 06:10; no product readings supplied; alarm and defrost entries coincide; duration and cause unconfirmed." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Imani's response in turn 10. It connects Noah's preceding point with Noah's follow-up. Food-safety lead is the assigned decision-maker for product safety under the site's procedures.

D Not here. This line belongs to turn 4, after Noah says: "That's all I've been given for temperature: the cabinet display. There are no food readings in the information I'm sending to the lead." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 96

Review 4 / Explanations / continued

4. Rebuild the exchange from "Report received, seasonal check still open". Ben: "Let me check the responsibilities: you own document correction and the Friday update. The seasonal-check owner and date are both still unconfirmed." Nia: [missing response] Ben: "I'll acknowledge receipt and keep the unresolved items separate. Friday goes in the update column, not the seasonal-test appointment column." Which response belongs in the gap?

B. That responsibility readback is accurate. It makes the commitment and both remaining scheduling questions visible instead of assigning the whole handover to one person by implication.

A Not here. This line belongs to turn 4, after Ben says: "Thanks for taking the document correction. I want to separate that from the outstanding check rather than mark everything complete on receipt of the report." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Nia's response in turn 18. It connects Ben's preceding point with Ben's follow-up. Responsibility readback distinguishes the assigned document task from the unassigned seasonal scheduling task.

C Not here. This line belongs to turn 6, after Ben says: "Who is arranging the seasonal check? I also can't find a booked date, so both responsibility and scheduling seem unresolved." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Ben says: "I nearly assigned that to you because you're handling the wrong sheet. That would add a responsibility you haven't actually accepted." It addresses that point, not the missing turn in the supplied exchange.

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Additional case 1 / Explanations

Extended conversation

1. sensible load

Turn 2. The supplied ten-kilowatt sensible load concerns temperature change; it must be compared with sensible capacity, not total capacity alone.

2. latent load

Turn 4. The two-kilowatt latent load concerns the moisture-removal part of this cooling duty; it is not a temperature value in degrees.

3. total capacity

Turn 6. Twelve kilowatts total equals nine sensible plus three latent here; extra latent capacity does not erase the stated sensible-capacity shortfall.

4. sensible heat ratio

Turn 8. SHR is sensible capacity divided by total capacity, nine divided by twelve or 0.75 for A; it is not output divided by electrical input.

5. comparison basis

Turn 10. The case gives consistent net capacities at the same conditions; B's 10.4 sensible and 2.6 latent exceed the supplied ten and two loads respectively.

6. load profile

Turn 12. Different combinations of temperature and moisture loads require different comparisons; the highest sensible fraction is not automatically best for every space.

Return to the conversation, page 99

Transfer exchange

1. 1.6

Manager: C's latent capacity is 1.6 kW.

Turn 1. Eight minus 6.4 equals 1.6 kW latent, below the supplied two-kilowatt latent load.

Additional case 1 / Explanations / continued

2. 0.4

Estimator: C therefore has a latent shortfall of 0.4 kW.

Turn 2. Two minus 1.6 equals a 0.4 kW shortfall despite C matching the eight-kilowatt total load.

3. D

Manager: Option D covers both supplied component loads numerically.

Turn 3. D supplies 6.3 sensible and 2.7 latent, above six and two respectively; this is not installation approval.

4. 0.7

Estimator: D's sensible heat ratio is 0.7 .

Turn 4. 6.3 divided by nine equals 0.7; SHR is a capacity split, not electrical efficiency.

Return to the conversation, page 102

Additional case 2 / Explanations

Extended conversation

1. total supply

Turn 2. Six hundred cubic meters per hour divided by a three-hundred-cubic-meter room gives two total-supply air changes per hour, including recirculated air.

2. outdoor-air flow

Turn 4. Only 150 cubic meters per hour is outdoor air in this case; the other 450 is recirculated, not an additional outdoor-air contribution.

3. air changes per hour

Turn 6. ACH divides a consistently expressed hourly airflow by room volume; the total-supply result is two and the outdoor-air result is 0.5.

4. outdoor-air fraction

Turn 8. Outdoor flow divided by total supply is 150/600 = 0.25, or 25 percent; this denominator is flow, not room volume.

5. time basis

Turn 10. A cubic meter contains one thousand liters and an hour has 3,600 seconds; 600 cubic meters per hour is about 166.7 liters per second.

6. air mixing

Turn 12. An air-change quantity describes delivered volume relative to room volume; it does not guarantee complete replacement of every portion of room air in a fixed interval.

Return to the conversation, page 104

Transfer exchange

1. three

Coordinator: Total-supply air changes per hour are three .

Turn 1. 720 divided by 240 equals three total-supply air changes per hour, including the outdoor portion.

Additional case 2 / Explanations / continued

2. 0.75

Specialist: Outdoor-air changes per hour are 0.75 .

Turn 2. 180 divided by 240 equals 0.75 outdoor-air changes per hour, not three.

3. 25

Coordinator: Outdoor air forms 25 percent of the supply.

Turn 3. 180 divided by 720 times one hundred equals twenty-five percent.

4. 50

Specialist: Outdoor flow is 50 liters per second.

Turn 4. 180 cubic meters per hour times one thousand divided by 3,600 equals fifty liters per second.

Return to the conversation, page 107

Additional case 3 / Explanations

Extended conversation

1. dew point

Turn 2. The supplied vapor-side dew point is 5 C; superheat is referenced to saturated vapor at the matching pressure, not the zero-degree bubble point.

2. superheat

Turn 4. Twelve degrees Celsius vapor temperature minus the five-degree dew point gives a seven-kelvin temperature difference; the original temperature reading remains twelve Celsius.

3. bubble point

Turn 6. The supplied liquid-side bubble point is 40 C; it is the saturated-liquid reference for subcooling at that reading's pressure.

4. subcooling

Turn 8. Forty degrees Celsius bubble point minus thirty-five degrees Celsius liquid temperature gives five kelvin subcooling; using dew would incorrectly give ten.

5. temperature glide

Turn 10. A blend with temperature glide has distinct bubble and dew temperatures at a given pressure; the supplied reference pairs differ by five kelvin.

6. matching pressure

Turn 12. The vapor and liquid readings have different pressure-specific reference pairs; a reference from the other location cannot be substituted into the calculation.

Return to the conversation, page 109

Transfer exchange

1. dew

Coordinator: The vapor calculation uses the dew point.

Turn 1. Superheat uses the saturated-vapor dew reference at the corresponding pressure, not the bubble value.

Additional case 3 / Explanations / continued

2. seven

Technician: The superheat is seven kelvin.

Turn 2. Eleven minus four equals seven kelvin; using minus one would incorrectly give twelve.

3. bubble

Coordinator: The liquid calculation uses the bubble point.

Turn 3. Subcooling uses the saturated-liquid bubble reference at the matching pressure, not the dew value.

4. six

Technician: The subcooling is six kelvin.

Turn 4. Thirty-seven minus thirty-one equals six kelvin; forty-two minus thirty-one would use the wrong reference.

Return to the conversation, page 112

Find a phrase by purpose

Clarifying comfort complaints and thermostat readings

Ask what the number means: Is 22 the setpoint or a measured room temperature?

Name the target: The thermostat is set to 22 degrees Celsius.

Preserve the missing reading: No confirmed ambient reading has been supplied.

Full phrase bank, page 7

Identifying equipment and linked components

Quote the request: The service request calls the system AC-4.

Quote the plan: The roof plan labels an outdoor unit CU-4.

Quote the register: The indoor register identifies air handler AH-4.

Full phrase bank, page 17

Explaining airflow and distribution reports

Name the two functions: S3 is supply air; R2 is return air.

Explain supply: Supply air is delivered into the room.

Explain return: Return air goes back toward the air-handling system.

Full phrase bank, page 29

Checking equipment ratings and refrigerant references

Ask the key question: Five kilowatts of which quantity?

Read proposal A: A lists 5 kilowatts of cooling capacity.

Read proposal B: B lists 5 kilowatts of electrical input.

Full phrase bank, page 40

Find a phrase by purpose / continued

Reporting moisture, symptoms, and service findings

State yesterday's account: The client reported water near AH-6 at 15:00 yesterday.

State today's observation: The visit note records the nearby floor as dry today.

Keep both times: Those accounts refer to different observations at different times.

Full phrase bank, page 53

Discussing refrigeration alarms and temperature records

Quote the actual record: The cabinet display recorded 8 degrees Celsius at 06:10.

Name the missing information: Product temperatures have not been supplied.

Keep the record type clear: That is a display value, not a verified food-temperature record.

Full phrase bank, page 63

Coordinating ductwork, access, and changes

Acknowledge the preference: I understand why you want the grille clear of the pendant light.

Name the request: You are asking to relocate the supply grille.

Avoid minimizing hidden work: The visible move may involve work above the ceiling.

Full phrase bank, page 76

Handing over commissioning and maintenance information

Acknowledge the report: You have received the commissioning report.

Name the exception: The report lists a deferred seasonal check.

Identify the document error: The maintenance sheet names the wrong model.

Full phrase bank, page 89

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References and publication notes

All cases, organizations, figures, and conversations are fictional teaching material, not transcripts. Communication practice does not replace qualified advice or current workplace procedures.

[1] US Department of Energy. Weatherization Training Glossary.

Background terminology for air distribution, temperature, humidity, and building systems. Historical program rules and technical procedures are not taught here. Source checked 10 October 2026.

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[3] US Environmental Protection Agency. Section 608 Technician Certification.

Context for professional boundaries around refrigerant work. The book does not teach certification requirements or authorize servicing. Source checked 10 October 2026.

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[4] US Food and Drug Administration. Refrigerator Thermometers: Cold Facts about Food Safety.

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Companion course

The website offers the eight-lesson course and related practice. This book can be used independently. No audio access or external account is needed for its exercises.

https://englishladder.com/efsp-hvac-refrigeration.html

Clear readings. Reliable handovers.

Eight conversations develop precise reporting and coordination: clarify a thermostat value, reconcile equipment labels, explain supply and return air, compare ratings, report moisture, escalate a cabinet alarm, discuss a grille move, and hand over unfinished commissioning items.

Clarifying comfort complaints and thermostat readings

Twenty-two degrees Celsius is the setpoint in this case. It must not be relabeled as ambient temperature. The complaint concerns staff feeling warm in the meeting room after 14:00, not a verified temperature or fault.

Identifying equipment and linked components

AC-4 appears in the request, CU-4 on the roof plan, and AH-4 in the indoor register. Keep all three with their sources. A shared numeral does not confirm a component pairing or served area.

Explaining airflow and distribution reports

S3 is labeled supply and R2 return on the current drawing. This tells the reader their stated functions. It does not establish actual flow rate, velocity, balance, or the cause of the discomfort.

Checking equipment ratings and refrigerant references

Both proposals state 5 kilowatts, but A describes cooling output and B electrical input. The unit can express either quantity. It does not follow that the systems have equal cooling capacities or equal efficiencies.

Phil Stilwell

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